CHAPTER 04

Ratio, Proportion & Variation

The basic applications of the concepts involved in this chapter are comparisons of two or more quantities and changes in their magnitudes, *e.g.*, comparison of the ages, weights income, savings, heights, volume, density, temperature etc. So this chapter is very useful in solving the problems of Data interpretation. Also each and every year one or two problems from this chapter is/are asked in CAT, either directly or application based for QA section. Last but not the least, the concepts of ratio, proportion and variations are very useful in solving most of the arithmetic problems. That's why these problems are usually asked in most of the competitive exams like CAT, XAT, IIFT, CMAT, GMAT, SSC, CGL and Bank PO.

Chapter Checklist

- Ratio
- Proportion
- Variation
- Problems Based on Ages
- Partnership
- CAT Test

4.1 Ratio

The comparison between two quantities in terms of magnitude is called the ratio, *i.e.*, it tells us that the one quantity is how many times the other quantity.

For example, Amit has 5 pens and Sarita has 3 pens. It means the ratio of number of pens between Amit and Sarita is 5 is to 3. It can be expressed as '5 : 3'.

NOTE It should be noted that in a ratio, the order of the terms is very important. For example, in the above illustration the required ratio is 5 : 3 while 3 : 5 is wrong.

So the ratio of any two quantities is expressed as $\frac{a}{b}$ or $a : b$.

The numerator 'a' is called the **antecedent** and denominator 'b' is called as **consequent**.

Rule of Ratio

The comparison of two quantities is meaningless if they are not of the same kind or in the same units (of length, volume or currency etc). We do not compare 8 boys and 6 cows or 15 litres and 5 toys or 5 metres and 25 centimetres. Therefore, to find the ratio of two quantities (of the same kind), it is necessary to express them in same units.

NOTE

1. We do not compare 8 boys and 6 cows, but we can compare the number (8) of boys and number (6) of cows. Similarly, we cannot compare the number (15) of litres and the number (5) of toys etc.
2. Ratio has no units.

Properties of Ratios

1. The value of a ratio does not change when the numerator and denominator both are multiplied by same quantities

i.e., $\frac{a}{b}=\frac{ka}{kb}=\frac{la}{lb}=\frac{ma}{mb}$ etc.

e.g., $\frac{3}{4}=\frac{6}{8}=\frac{9}{12}\ldots$ etc. have the same ratio.

2. The value of a ratio does not alter (or change) when the numerator and denominator both are divided by same quantities *i.e.*, $\frac{a}{b}=\frac{a/k}{b/k}=\frac{a/l}{b/l}=\frac{a/m}{b/m}$ etc.

e.g., $\frac{3}{4}=\frac{3/2}{4/2}=\frac{3/3}{4/3}=\frac{3/4}{4/4}\ldots$ etc are same ratio.

3. The ratio of two fractions can be expressed in ratio of integers. *e.g.*, $\frac{3/4}{5/4}=\frac{3}{4}\times\frac{4}{5}=\frac{3}{5}$

4. When two or more than two ratios are multiplied with each other, then it is called as compounded ratio *e.g.*,

$\frac{2}{3}\times\frac{4}{5}\times\frac{6}{7}=\frac{16}{35}$ is the compounded ratio of $\frac{2}{3},\frac{4}{5},\frac{6}{7}$

So, $\frac{a}{b}\times\frac{c}{d}\times\frac{e}{f}\ldots=\frac{k}{l}$ (compounded ratio)

5. When the ratio is compounded with itself, it is called as duplicate, triplicate ratios etc. *e.g.*,

$\frac{a}{b}\times\frac{a}{b}=\frac{a^2}{b^2}=\left(\frac{a}{b}\right)^2$ is called as duplicate ratio of $\frac{a}{b}$ and

$\frac{a}{b}\times\frac{a}{b}\times\frac{a}{b}=\left(\frac{a}{b}\right)^3$ is called as triplicate ratio of $\frac{a}{b}$.

Similarly, $\sqrt{\left(\frac{a}{b}\right)}=\left(\frac{a}{b}\right)^{1/2}$ is called as sub-duplicate ratio and $\sqrt[3]{\left(\frac{a}{b}\right)}=\left(\frac{a}{b}\right)^{1/3}$ is called as sub-triplicate ratio of $\frac{a}{b}$.

6. $\frac{a}{b}=\frac{c+am}{d+bm}$ if and only if $\frac{c}{d}=\frac{a}{b}$

i.e., $\frac{30}{40}=\frac{30+3m}{40+4m}=\frac{33}{44}=\frac{36}{48}\ldots$ etc at $m=1,2,\ldots$

This property is very useful when we compare two fractions *e.g.*, to compare between $\frac{10}{20}$ and $\frac{12}{22}$

We see that $10/20=1/2$

Now if $\frac{10}{20}$ and $\frac{12}{22}$ are equal then $\frac{12}{22}=\frac{10+m}{22+2m}$

Now putting $m=2$, we don't get the required fraction.

So $\frac{12}{22}\neq\frac{10}{20}$ for any value of m, $\frac{12}{22}\neq\frac{10}{20}$ (in terms of ratio)

7. $\frac{a+k}{b+k}<\frac{a}{b}$ if for every positive k, $\frac{a}{b}>1$ and $\frac{a-k}{b-k}>\frac{a}{b}$

8. $\frac{a+k}{b+k}>\frac{a}{b}$ if for every positive k, $\frac{a}{b}<1$ and $\frac{a-k}{b-k}<\frac{a}{b}$

9. $\frac{a+c}{b+d}>\frac{a}{b}$ if $\frac{c}{d}>\frac{a}{b}$ **10.** $\frac{a+c}{b+d}<\frac{a}{b}$ if $\frac{c}{d}<\frac{a}{b}$

11. If $\frac{a}{b}=\frac{c}{d}=\frac{e}{f}=\frac{g}{h}=\ldots=k$ then $\frac{a+c+e+g+\ldots}{b+d+f+h+\ldots}=k$

12. Let $\frac{a}{b},\frac{c}{d},\frac{e}{f},\frac{g}{h}\ldots$ be some different ratios, then the value of $\left(\frac{a+c+e+g+\ldots}{b+d+f+h+\ldots}\right)$ must lies between the lowest and highest ratios.

13. If $a:b$ and $b:c$ are given, then

$$a:b:c=\overset{a}{\underset{b}{\downarrow}}\swarrow\overset{b}{\underset{c}{\downarrow}}=(a\cdot b):(b\cdot b):(b\cdot c)$$

14. If the ratios between $a:b, b:c, c:d, d:e$ are given individually, then the combined ratio of $a:b:c:d:e$.

$$\begin{matrix}a:b\\b:c\\c:d\\d:e\end{matrix}\quad\begin{matrix}a\\b\\c\\d\downarrow\end{matrix}\;:\;\begin{matrix}b\\b\\c\\d\downarrow\end{matrix}\;:\;\begin{matrix}b\\c\\c\\d\downarrow\end{matrix}\;:\;\begin{matrix}b\\c\\d\\d\downarrow\end{matrix}\;:\;\begin{matrix}b\\c\\d\\e\downarrow\end{matrix}$$

i.e., $a:b:c:d:e=(a.b.c.d):(b.b.c.d)$
$:(b.c.c.d):(b.c.d.d):(b.c.d.e)$

Exp. 1) Find the ratio of 25 to 40.

Solution $\frac{25}{40}=\frac{5\times5}{5\times8}=\frac{5}{8}$

NOTE To get the ratio, we rationalize the fractions by cancelling out the common factors of numerator and denominator.

Exp. 2) Find the ratio of 90 cm to 1.5 m.

Solution 1.5 m = 150 cm (units must be same)

So, the required ratio $=\frac{90}{150}=\frac{3}{5}$

Exp. 3) The number of boys and girls in a school are 576 and 480 respectively. Express the ratio of the number of boys to that of girls in the simplest form.

Solution Required ratio $=\frac{576}{480}=\frac{6}{5}$ $\left(\because\frac{576}{480}=\frac{96\times6}{96\times5}\right)$

Exp. 4) Shukla earns ₹ 14000 per month and Mishra earns ₹ 18000 per month. Find the ratio of Shukla's salary to Mishra's salary.

Solution Required ratio $=\frac{14{,}000}{18{,}000}=\frac{7}{9}$

Exp. 5) Out of 144 persons working in an office, 56 are men and the remaining are women. Find the ratio of number of women to number of men.

Solution Ratio $= \frac{88}{56} = \frac{11}{7}$ ($\because$ number of women $= 144 - 56$)

Exp. 6) In a club having 100 members, 20 play carrom, 24 play table-tennis and 16 play cricket and the remaining members do not play any game. No member plays more than one game. Find the ratio of the number of members who play.

(a) Carrom to the number of those who play table-tennis.
(b) Cricket to the number of those who play carrom.
(c) Cricket to the number of those who do not play any game.
(d) Table-tennis to the number of those who do not play any game.
(e) Some game to the number of those who do not play any game.

Solution Total members = 100

Carrom = 20
Table-tennis = 24
Cricket = 16
No any game = 40

(a) $\frac{20}{24} = \frac{5}{6}$ (b) $\frac{16}{20} = \frac{4}{5}$ (c) $\frac{16}{40} = \frac{2}{5}$ (d) $\frac{24}{40} = \frac{3}{5}$
(e) $\frac{60}{40} = \frac{3}{2}$

Exp. 7) A person earns ₹ 1200 per day and spends ₹ 800. Find the ratio of his savings to expenditure.

Solution $\frac{400}{800} = \frac{1}{2}$ (Savings = Income – Expenditure)
$= 1 : 2$

Exp. 8) Simplify the following ratios :

(a) $\frac{1}{6} : \frac{1}{8}$ (b) $2\frac{1}{3} : 3\frac{1}{2}$ (c) $\frac{5}{6} : \frac{3}{8} : 3\frac{3}{4}$

Solution (a) $\frac{1}{6} : \frac{1}{8} = \frac{1/6}{1/8} = \frac{1}{6} \times \frac{8}{1} = \frac{4}{3}$ or 4 : 3

Alternatively $\frac{1}{6} : \frac{1}{8} = \frac{48}{6} : \frac{48}{8}$ $\left(\text{or } \frac{24}{6} : \frac{24}{8}\right)$
$= 8 : 6 = 4 : 3$

NOTE In case of fractions, convert them to whole numbers by multiplying each term by the LCM of their denominators.

(b) $2\frac{1}{3} : 3\frac{1}{2} = \frac{7}{3} : \frac{7}{2} = \frac{7/3}{7/2} = \frac{2}{3}$ or 2 : 3

NOTE In case of two fractions if numerators are same, then the required ratio is the inverse ratio of the fractions.

(c) $\frac{5}{6} : \frac{3}{8} : 3\frac{3}{4} = \frac{5}{6} : \frac{3}{8} : \frac{15}{4} = \frac{5}{6} \times 24 : \frac{3}{8} \times 24 : \frac{15}{4} \times 24$
$= 20 : 9 : 90$ {$\because$ LCM of 6, 8 and 4 is 24}

Exp. 9) Divide 14 toffees among Ankita and Anshul in the ratio 5 : 2.

Solution According to the question, if there are 7 toffees then Ankita will have 5 and Anshul will have 2 toffees but since there are 14 toffees, which is twice of 7. So Ankita will have 10 toffees and Anshul will have 4 toffees.

Shortcut

Ankita		Anshul
5	:	2
$14 \times \frac{5}{(5+2)}$	:	$14 \times \frac{2}{(5+2)}$
10	:	4

Exp. 10) Three boys are aged 2 years, 4 years and 8 years. They want to divide seventy rupees in the ratio of their ages. How much money would each get?

Solution The ratio of their ages = $A : B : C = 2 : 4 : 8 = 1 : 2 : 4$

A	:	B	:	C
1	:	2	:	4
$70 \times \frac{1}{1+2+4}$	:	$70 \times \frac{2}{1+2+4}$	:	$70 \times \frac{4}{1+2+4}$
10	:	20	:	40

So A gets ₹ 10, B gets ₹ 20 and C gets ₹ 40 if the ratio of their ages is 2 : 4 : 8.

Exp. 11) An amount of ₹ 100 is being divided among two persons in the ratio $\frac{1}{10} : \frac{1}{15}$. How much money does each get?

Solution $\frac{1}{10} : \frac{1}{15} = \frac{1}{10} \times 30 : \frac{1}{15} \times 30 = 3 : 2$ (here, 30 is the LCM of 10 and 15). So, the ratio of amount of money
$= \frac{3}{5} \times 100 : \frac{2}{5} \times 100 = 60$ and 40

Exp. 12) The lengths of sides of a triangle are in the ratio 2 : 3 : 4. If the perimeter of the triangle is 63 cm, find the lengths of the sides of the triangle.

Solution Let the sides of triangle be $2x$, $3x$ and $4x$, then
$2x + 3x + 4x = 63 \Rightarrow 9x = 63 \Rightarrow x = 7$
$\therefore$ The sides of triangle $= 2x, 3x, 4x = 14, 21, 28$

Alternatively $63 \times \frac{2}{(2+3+4)}$ and $63 \times \frac{3}{(2+3+4)}$
and $63 \times \frac{4}{(2+3+4)} = 14, 21$ and 28.

Exp. 13) Divide 1224 into three parts such that first part be double that of second part and second part be $\frac{1}{3}$ of the third part.

Solution Let A, B and C be three parts respectively, then
$A : B = 2 : 1$ and $B : C = 1 : 3$
$\therefore$ $A : B : C = 2 : 1 : 3$ $\therefore$ $A = 1224 \times \frac{2}{6} = 408$
$B = 1224 \times \frac{1}{6} = 204 \Rightarrow C = 1224 \times \frac{3}{6} = 612$

Exp. 14) If $A:B=3:4$, $B:C=5:2$ then find the value of $A:B:C$.

Solution $A:B=3:4$

$$\begin{bmatrix} 3 & : & 4 \\ \downarrow & \nearrow & \downarrow \\ 5 & : & 2 \end{bmatrix}$$

and $B:C=5:2$

$$A:B:C=(3\times5):(4\times5):(4\times2)$$

or $A:B:C=15:20:8$

Exp. 15) The ratio of $A:B=1:3$, $B:C=2:5$, $C:D=2:3$. Find the value of $A:B:C:D$.

Solution $A:B=1:3$

$B:C=2:5$

$C:D=2:3$

$$A:B:C:D=(1\times2\times2):(3\times2\times2):(3\times5\times2):(3\times5\times3)$$

$$A:B:C:D=4:12:30:45$$

Exp. 16) There are two types of mixtures of milk and water. In the first mixture, out of 12 litres of mixture, 5 litre is milk only and in the second mixture, 6 litre is milk and 12 litre is water. Which one mixture is better in terms of milk's strength?

Solution

First Mixture	Second Mixture	
$\frac{5}{12}$	$\frac{6}{18}$	$(6+12=18)$ (milk + water = mixture)
$\frac{5}{12}\times36$	$\frac{6}{18}\times36$	
15	12	

So the first mixture has more milk in comparison to water.

Exp. 17) The ratio of salary of $A:B=1:2$, $B:C=3:4$, $C:D=5:6$ and $D:E=7:8$. What is the ratio of salary of A and E?

Solution

$A:B$	$1:2$
$B:C$	$3:4$
$C:D$	$5:6$
$D:E$	$7:8$

$$A:B:C:D:E=(1\times3\times5\times7):(2\times3\times5\times7):(2\times4\times5\times7):(2\times4\times6\times7):(2\times4\times6\times8)$$

or $A:B:C:D:E=105:210:280:336:384$

So the ratio of salary of $A:E=105:384=35:128$

NOTE In every next step, we leave the left term and adopt right term.

Exp. 18) If $\frac{a}{b}=\frac{3}{4}$, then find the value of $7a-4b:3a+b$.

(a) 7 : 1 (b) 5 : 13
(c) 12 : 1 (d) none of these

Solution Simply substitute the value of a and b as 3 and 4 in the given algebraic ratio

as $(7\times3-4\times4):(3\times3+4)=5:13$

Exp. 19) If $a:b=3:2$ and $b:c=6:5$ then $a:b:c$ is equal to

(a) 9 : 6 : 5 (b) 9 : 6 : 10 (c) 3 : 3 : 5 (d) 3 : 6 : 5

Solution $a:b=3:2$

$b:c=6:5$

$$a:b:c=(3\times6):(2\times6):(2\times5)=18:12:10=9:6:5$$

Exp. 20) The sum of two natural numbers is 64. Which of the following cannot be the ratio of these two numbers?

(a) 3 : 5 *(b)* 1 : 3 *(c)* 7 : 9 *(d)* 3 : 4

Solution Let the numbers be $3x$ and $5x$, then

$$3x+5x=64 \Rightarrow 8x=64$$

$\Rightarrow x=8$, which is possible

The numbers are 24, 40.

Take another option: $x+3x=64 \Rightarrow x=16$

The numbers are 16 and 48.

Check for option (c) :

$$7x+9x=64 \Rightarrow 16x=64 \Rightarrow x=4$$

The numbers are 28 and 36.

Check for option (d)

$$3x+4x=64 \Rightarrow 7x=64$$

$$x=\frac{64}{7}$$

Numbers are $\frac{3\times64}{7}$ and $\frac{4\times64}{7}$ or $\frac{192}{7}$ and $\frac{256}{7}$

Which are not the natural numbers.

Hence option (d) is the required answer.

Exp. 21) Monthly incomes of A and B are in the ratio of 4 : 3 and their savings are in the ratio of 3 : 2. If the expenditure of each will be ₹ 600, then the monthly incomes of each are :

(a) 1800, 2400 (b) 2400, 1600 (c) 2400, 1800 (d) 1600, 1200

Solution Income = Exp. + Savings

$A \rightarrow 4x=3y+600$

$B \rightarrow 3x=2y+600$

Therefore, $4x-3y=600$ and $3x-2y=600$

$\Rightarrow 4x-3y=3x-2y \Rightarrow x=y$

$\therefore 4x-3x=600 \Rightarrow x=600$

Then, the income of $A=4\times600=2400$

and income of $B=3\times600=1800$

Alternatively Check the options. Consider (c)

Income	A		B	
	2400		1800	
	(4)	:	(3)	$\Rightarrow$ correct
Again,	2400		1800	
	$-\ 600$		$-\ 600$	
Savings	1800		1200	
	(3)	:	(2)	$\Rightarrow$ correct

Hence, option (c) is correct. If you check other options, the ratio will not match.

Exp. 22) A, B and C have 40, x and y balls with them respectively. If B gives 20 balls to A, he is left with half as many balls as C. If together they had 60 more balls, each of them would have had 100 balls on an average. What is value of $x : y$?

(a) 3 : 2 *(b)* 4 : 6 *(c)* 2 : 1 *(d)* 3 : 4

Solution From the last statement

$$\frac{100 + x + y}{3} = 100$$

$$\Rightarrow \quad x + y = 200 \qquad \ldots(i)$$

Again from first statement, $\frac{x - 20}{y} = \frac{1}{2}$

$$\Rightarrow \quad 2x - y = 40 \qquad \ldots(ii)$$

an solving eqs. (i) and (ii), we get

$$x = 80 \quad \text{and} \quad y = 120$$

Therefore, required ratio of $x : y = 2 : 3$

Hence, option (b) is correct.

Alternatively After forming the eqs. (i), we can go through the options. Let us assume option (b).

$$x + y = 200$$

$$\because \quad x : y = 4 : 6$$

$$\Rightarrow \quad x = 80 \quad \text{and} \quad y = 120$$

Now from the first statement, $\frac{x - 20}{y} = \frac{1}{2}$

So $\frac{80 - 20}{120} = \frac{1}{2}$ (verified)

Hence option (b) is correct.

Exp. 23) The incomes of A, B and C are in the ratio of 12 : 9 : 7 and their spendings are in the ratio 15 : 9 : 8. If A saves 25% of his income. What is the ratio of the savings of A, B and C?

Solution Income = Expenditure + Saving

$A \rightarrow \quad 12x = 15y + 3x \qquad (3x = 25\% \text{ of } 12x)$

$B \rightarrow \quad 9x = 9y + (9x - 9y)$

$C \rightarrow \quad 7x = 8y + (7x - 8y)$

Therefore, $12x - 3x = 15y$

$$\Rightarrow \quad \frac{x}{y} = \frac{5}{3} \quad \Rightarrow \quad y = \frac{3x}{5}$$

Therefore, savings = (income – expenditure)

$$A = 12x - 9x = 3x$$

$$B = 9x - \frac{27}{5}x = \frac{18}{5}x$$

$$C = 7x - \frac{24}{5}x = \frac{11}{5}x$$

i.e., the ratio of savings of $A : B : C = 3x : \frac{18}{5}x : \frac{11}{5}x$

$$= 15x : 18x : 11x = 15 : 18 : 11$$

Exp. 24) There are total 100 coins consisting of 20 paise, 50 paise and ₹ 1 in the ratio of 7 : 8 : 5. What is the no. of coins of 50 paise if the difference between the amount yielded by 20 paise and ₹ 1 coin is 18?

(a) 32 *(b)* 40

(c) 26 *(d)* 56

Solution $(20 \times 7x) \sim (100 \times 5x) = 1800$ (₹ 18 = 1800 paise)

$$(140x) \sim (500x) = 1800$$

$$\Rightarrow \quad 360x = 1800$$

$$\Rightarrow \quad x = 5$$

Therefore, number of coins of 50 paise $= 8 \times 5$

$$= 40$$

Alternatively Let there be 40 coins of 50 paise denomination, then

The no. of coins of A, B and $C = 7x : 8x : 5x$

$$= 35 : 40 : 25$$

Therefore, the amount from 20 paise coins

$$= 35 \times 20 = ₹ 7$$

and amount from ₹ 1 coins $= 1 \times 25 =$ ₹ 25

Hence difference = ₹ 18 (25 – 7)

Thus the presumed option is correct.

Exp. 25) There are 43800 students in 4 schools of a city. If half of the first, two-third of the second, three-fourth of the third and four-fifth of the fourth are the same number of students, then find the ratio of number of students of A and D if A, B, C and D be the first, second, third and fourth schools respectively.

Solution $$\frac{A}{2} = \frac{2B}{3} = \frac{3C}{4} = \frac{4D}{5}$$

Therefore, $\frac{A}{B} = \frac{4}{3}$ and $\frac{B}{C} = \frac{9}{8}$ and $\frac{C}{D} = \frac{16}{15}$

$$\therefore \quad A : B = 4 : 3$$

$$B : C = 9 : 8$$

$$C : D = 16 : 15$$

$$\therefore \quad A : B : C : D = (4 \times 9 \times 16) : (3 \times 9 \times 16) : (3 \times 8 \times 16) : (3 \times 8 \times 15)$$

$$A : B : C : D = 576 : 432 : 384 : 360$$

Therefore, the ratio of number of students of

$$A \text{ and } D = 576 : 360 = 8 : 5$$

Alternatively $$\frac{A}{2} = \frac{2}{3}B = \frac{3}{4}C = \frac{4}{5}D = k$$

then $$A : B : C : D = 2k : \frac{3}{2}k : \frac{4}{3}k : \frac{5}{4}k$$

$$= 24k : 18k : 16k : 15k$$

$$\therefore \quad A : B : C : D = 24 : 18 : 16 : 15$$

$$\therefore \quad A : D = 24 : 15 = 8 : 5$$

4.2 Proportion

An equality of two ratios is called a proportion and we say that four numbers are in proportion.

That is, if $\frac{a}{b}=\frac{c}{d}$, or $a:b=c:d$, then we say that a, b, c and d are in proportions and we write them as $a:b::c:d$, where the symbol '::' indicates proportion and it is read as 'a is to b as c is to d'.

Here a and d are called **extremes** (or extreme terms) and b and c are called as **means** (or middle terms). Thus four numbers are said to be in proportion, if the ratio of the first to the second number is equal to the ratio of the third to the fourth number. For Example, $2:3::4:6$.

Proportionality Test

If four numbers (quantities) are in proportion, then product of the extremes is equal to the product of the means and if these are not in proportion, then product of extremes is not equal to the product of the means.

That is if $a:b::c:d$, then $a\times d=b\times c$

Thus it is clear that if three out of four terms of a proportion are given, we can find the fourth term by using this proportionality test.

Proportionality Theorems

(a) **Invertendo:** If $\frac{a}{b}=\frac{c}{d}\Rightarrow\frac{b}{a}=\frac{d}{c}$

(b) **Alternando:** If $\frac{a}{b}=\frac{c}{d}\Rightarrow\frac{a}{c}=\frac{b}{d}$

(c) **Componendo:** If $\frac{a}{b}=\frac{c}{d}\Rightarrow\left(\frac{a+b}{b}\right)=\left(\frac{c+d}{d}\right)$

(d) **Dividendo:** If $\frac{a}{b}=\frac{c}{d}\Rightarrow\left(\frac{a-b}{b}\right)=\left(\frac{c-d}{d}\right)$

(e) **Componendo and Dividendo:** If $\frac{a}{b}=\frac{c}{d}$

$\Rightarrow\quad\left(\frac{a+b}{a-b}\right)=\left(\frac{c+d}{c-d}\right)$

NOTE

1. Each of the two results (a) and (b) can be obtained by cross product.
2. The result (e) can be obtained by dividing result (c) by result (d).

Continued Proportion

If a, b and c are three numbers such that $a:b=b:c$, then these numbers a, b and c are said to be in **continued proportion** or simply in proportion.

That is, if $a:b=b:c$ then $b^2=ac$

Here b is said to be the **mean proportional** to a and c, and c is said to be the **third proportional** to a and b.
For Example, $3:9::9:27$

Here, $9\times9=3\times27$. That means 3, 9 and 27 are in continued proportion.

NOTE Sometimes the above idea is also expressed by saying that the three numbers are in the ratio 3:9:27. Thus, if three quantities are proportionals, the first is to the third as the duplicate ratio of the first to the second. *i.e.*, if $a:b=b:c$, then $a:c=a^2:b^2$

$$\left[\because\frac{a}{b}=\frac{b}{c}\Rightarrow\frac{a^2}{b^2}=\frac{ab}{bc}\Rightarrow\frac{a^2}{b^2}=\frac{a}{c}\Rightarrow a:c=a^2:b^2\right]$$

Continued Proportionality Theorems

If a, b and c are three numbers such that $a:b=b:c$, that is $\frac{a}{b}=\frac{b}{c}$ then,

(a) $b^2=ac$

Proof $\frac{a}{b}=\frac{b}{c}\Rightarrow b^2=ac$

(b) $\frac{b}{a}=\frac{c}{b}$

Proof $\frac{a}{b}=\frac{b}{c}\Rightarrow b^2=ac\Rightarrow\frac{b}{a}=\frac{c}{b}$

(c) $\frac{a}{c}=\frac{a^2}{b^2}$ (Duplicate Ratio)

Proof $\frac{a}{b}=\frac{b}{c}\Rightarrow\frac{a}{b}\times\frac{a}{b}=\frac{b}{c}\times\frac{a}{b}\Rightarrow\frac{a^2}{b^2}=\frac{a}{c}$

(d) $\frac{a}{c}=\frac{b^2}{c^2}$ (Duplicate Ratio)

Proof $\frac{a}{b}=\frac{b}{c}\Rightarrow\frac{a}{b}\times\frac{b}{c}=\frac{b}{c}\times\frac{b}{c}\Rightarrow\frac{a}{c}=\frac{b^2}{c^2}$

(e) $\frac{a}{c}=\frac{a^2+b^2}{b^2+c^2}$

Proof Since $\frac{a}{c}=\frac{a^2}{b^2}$ and $\frac{a}{c}=\frac{b^2}{c^2}$

$\Rightarrow\quad\frac{a+a}{c+c}=\frac{a^2+b^2}{b^2+c^2}\Rightarrow\frac{a}{c}=\frac{a^2+b^2}{b^2+c^2}$

Exp. 1) The first, second and fourth terms of a proportion are 5, 15 and 90 respectively. Find the third term.

Solution Let the third term be x, then $5, 15, x, 90$ are in proportion

i.e., $5:15::x:90$

$\Rightarrow\quad 5\times90=15\times x$ (by the Ist property)

$\therefore\quad x=30$

Exp. 2) The ratio of length to width of a rectangular sheet of paper is 5 : 3. If the width of the sheet is 18 cm, find its length.

Solution Let the length of sheet of paper be x cm. Then the ratio of length to width = $x : 18$

Thus $x : 18 = 5 : 3 \Rightarrow x \times 3 = 18 \times 5 \Rightarrow x = 30$

Hence the length of paper = 30 cm.

Exp. 3) If 81, x, x, 256 are in proportion, find x.

Solution $81 : x :: x : 256 \Rightarrow 81 \times 256 = x \times x \Rightarrow x = 144$

Exp. 4) The ratio between the number of men and women in an office is 5 : 7.
If the number of women working in the office is 56, find the number of men working in the office.

Solution $5 : 7 = x : 56$ (suppose number of men = x)

$\Rightarrow$ $x = 40$ (by the first property)

Therefore, number of men in the office = 40

Exp. 5) The age of Chandi and Radhika are in the ratio 5 : 3. If Chandi's age is 20 years, find the age of Radhika.

Solution $5 : 3 = 20 : x$

$\Rightarrow$ $x = \dfrac{3 \times 20}{5}$ (by the first property)

$\Rightarrow$ $x = 12$

Hence the age of Radhika is 12 years.

Exp. 6) The ratio of the number of boys to that of girls in a school is 9 : 11. If the number of girls in the school is 2035, find :

(a) number of boys in school

(b) number of students in school.

Solution (a) $9 : 11 = x : 2035 \Rightarrow x = \dfrac{9 \times 2035}{11}$

$x = 1665$ (number of boys)

(b) number of students

= number of boys + number of girls = 3700

Exp. 7) What is the least possible number which must be subtracted from 16, 19 and 23 so that the resulting numbers are in continued proportion?

(a) 2 (b) 4 (c) 6 (d) 7

Solution Going through options, we find option (d) is correct.

Alternatively $(16 - x) : (19 - x) :: (19 - x) : (23 - x)$

$\Rightarrow$ $(19 - x)^2 = (16 - x)(23 - x)$

By solving the above equation, we get $x = 7$

Exp. 8) If $(a + b) : (a - b) = 15 : 1$, then the value of $a^2 - b^2$ is :

(a) 56 (b) 15 (c) 112 (d) 8

Solution $\dfrac{(a + b)}{(a - b)} = \dfrac{15}{1} \Rightarrow \dfrac{a}{b} = \dfrac{8}{7}$ (by componendo and dividendo)

Therefore $a^2 - b^2 = 64 - 49 = 15$

Exp. 9) The mean proportional between 8 and 98 is :

(a) 16 (b) 53 (c) 112 (d) 28

Solution $8 : x :: x : 98$

$x^2 = 8 \times 98 \Rightarrow x = 28$

Hence option (d) is correct.

Exp. 10) The students in three classes are in the ratio of 2 : 3 : 4. If 40 students are added in each class, the ratio becomes 4 : 5 : 6. Find the total number of students in all the three classes is :

(a) 270 (b) 180 (c) 126 (d) 135

Solution $2x + 40 = 4y$ …(i)

$3x + 40 = 5y$ …(ii)

$4x + 40 = 6y$ …(iii)

Therefore $2x = 40 \Rightarrow x = 20$

Hence, total number of students = $2x + 3x + 4x = 9x$

$= 9 \times 20 = 180$

Alternatively It can be solved through options also.

Exp. 11) The dimensions of a photograph are 4 and 1.8 cms. If the breadth of the enlarged photo is 4.5 cm and it was enlarged proportionally then what is the new length of new photograph?

(a) 6 (b) 5.4 (c) 10 (d) 9

Solution $4 : 1.8 = x : 4.5$

$\therefore$ $x = 10$

Thus, the length of new photograph is 10 cm.

Exp. 12) Two equal containers are filled with the mixture of milk and water. The concentration of milk in each of the containers is 20% and 25% respectively. What is the ratio of water in both the containers respectively?

(a) 15 : 16 (b) 16 : 15 (c) 4 : 5 (d) 5 : 4

Solution

Milk	20%	25%
Water	80%	75%

Therefore, required ratio = $\dfrac{80}{75} = \dfrac{16}{15}$ or 16 : 15

Exp. 13) A cat takes 7 steps for every 5 steps of a dog, but 5 steps of a dog are equal to 6 steps of cat. What is the ratio of speed of cat to that of dog?

(a) 24 : 25 (b) 42 : 25 (c) 24 : 19 (d) 25 : 42

Solution

	CAT	**DOG**
Given speed	7 steps	5 steps

But the length of 5 steps of dog = length of 6 steps of cat

It means the ratio of length covered by dog is to cat = $\dfrac{6}{5}$

Therefore, in each step a dog will cover $\dfrac{6}{5}$ times distance than that of a cat.

Thus the ratio of actual speed of cat is to dog

$= 7 : 5 \times \dfrac{6}{5} = 7 : 6$

Alternatively Actual speed of $A : B$

$$= \frac{\text{Given speed of } A}{\text{No. of steps of } A \text{ in terms of length}} : \frac{\text{Given speed of } B}{\text{No. of steps of } B \text{ in terms of length}}$$

$$= \text{CAT} : \text{DOG} \quad = \frac{7}{6} : \frac{5}{5} = 7 : 6$$

Exp. 14) A camel pursue an elephant and takes 5 leaps for every 7 leaps of the elephant, but 5 leaps of elephant are equal to 3 leaps of camel. What is the ratio of speeds of camel and elephant?

Solution Ratio of speed of camel and elephant

$$= \frac{5}{3} : \frac{7}{5} = \frac{5}{3} \times 15 : \frac{7}{5} \times 15 = 25 : 21$$

Introductory Exercise 4.1

1. If $A : B = 4 : 5$, $B : C = 3 : 4$, $C : D = 7 : 11$, then $A : D$ is :
(a) 3 : 4 (b) 21 : 55 (c) 21 : 44 (d) 7 : 5

2. Mean proportional between 17 and 68 is :
(a) 51 (b) 24 (c) 4 (d) 34

3. Third proportional between 16 and 36 is :
(a) 64 (b) 144 (c) 81 (d) 49

4. If $a : b = 2 : 3$, then $(5a + b) : (3a + 2b)$ is :
(a) 13 : 12 (b) 15 : 17 (c) 12 : 13 (d) 13 : 11

5. $a = 2b = 3c = 4d$, then $a : b : c : d$ is :
(a) 12 : 3 : 6 : 4 (b) 3 : 4 : 6 : 12
(c) 6 : 12 : 4 : 3 (d) 12 : 6 : 4 : 3

6. The fourth proportional to 4, 7 and 20 is :
(a) 28 (b) 21 (c) 18 (d) 35

7. If $\sqrt{2} : (1 + \sqrt{3}) :: \sqrt{6} : x$, then x is equal to :
(a) $1 + \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $\sqrt{3} + 3$ (d) $2\sqrt{3}$

8. If $\frac{a}{3} = \frac{b}{4} = \frac{c}{5}$ then $\frac{a + b + c}{b} = ?$
(a) 2 (b) 3 (c) 4 (d) 5

9. If $\frac{a}{b} = \frac{c}{d}$, then :
(a) $\frac{a + b}{a - b} = \frac{c + d}{c - d}$ (b) $\frac{a + b}{a^2} = \frac{c + d}{d^2}$
(c) $\frac{a + b}{a^2} = \frac{c + d}{c^2}$ (d) $ac = bd$

10. If $a : b = b : c = c : d$ then $\frac{a}{b}, \frac{b}{c}, \frac{c}{d}$ are :
(a) in AP
(b) in continued proportion
(c) in GP
(d) both (b) and (c)

11. If $(a + b) : (a - b) = 3 : 2$, the $(a^2 - b^2) : (a^2 + b^2)$ equals :
(a) 5 : 13 (b) 12 : 13
(c) 9 : 4 (d) none of these

12. Two whole numbers, whose sum is 64, cannot be in the ratio :
(a) 1 : 7 (b) 3 : 5 (c) 5 : 11 (d) 1 : 2

13. Two numbers are in the ratio 3 : 4. The difference between their squares is 28. Find the greater number
(a) 12 (b) 8 (c) 24 (d) 16

14. If a, b, c, d, e, f and g are in continued proportion, then the value of $a.b.c.e.f.g$ is :
(a) d^3 (b) d^7
(c) d^6 (d) none of these

15. If A and B shared ₹ 1300 in the ratio 1 : 12, how much did A get?
(a) 120 (b) 1200 (c) 100 (d) 1000

16. 3600 are divided among A, B and C so that half of A's part, one-third of B's part and one-sixth of C's part are equal. Then C's part is :
(a) 720 (b) 2160 (c) 1080 (d) 810

17. A sum of ₹ 21000 is divided among A, B and C such that shares of A and B are in the ratio of 2 : 3 and those of B and C are in the ratio 4 : 5. The amount received by A is :
(a) ₹ 6000 (b) ₹ 4500 (c) ₹ 4800 (d) ₹ 8400

18. A certain amount was divided between A and B in the ratio 7 : 9. If B's share was ₹ 7200, the total amount was :
(a) ₹ 1280 (b) ₹ 6300 (c) ₹ 5600 (d) ₹ 12800

19. ₹ 11250 are divided among A, B and C so that A may receive one-half as much as B and C together receive and B receives one-fourth of what A and C together receive. The share of A is more than that of B by :
(a) ₹ 2500 (b) ₹ 1500 (c) ₹ 1800 (d) ₹ 650

20. A girl 1.2 metre tall casts a shadow 1.1 m at the time when a building casts a shadow 6.6 m long. The height of the building is :
(a) 2.7 m (b) 7.2 m (c) 6.0 m (d) 5.5 m

21. The prices of Bajaj Scooter and Bajaj Pulser are in the ratio of 4 : 9. If the Bajaj Pulser costs ₹ 30000 more than a Bajaj Scooter, the price of Bajaj Pulser is :
(a) ₹ 63000 (b) ₹ 45000
(c) ₹ 54000 (d) ₹ 60000

22. What is the ratio whose terms differ by 40 and the measure of which is 2/7?
(a) 16 : 56 (b) 14 : 49
(c) 15 : 36 (d) 16 : 72

23. Two numbers are in the ratio 3 : 5. If 9 be subtracted from each, then they are in the ratio of 12 : 23. The second number is :
(a) 53 (b) 54
(c) 55 (d) 52

24. Consider the following statements :
(1) If both the terms of a ratio are multiplied or divided by the same natural number, then the ratio remains unaltered.
(2) A statement which states that two ratios are equivalent is called proportion.
(3) If 4 quantities are in proportion, the product of extremes is not equal to the means.
(4) The mean proportion between any two numbers is equal to the square root of their product. The wrong one statements is/are :
(a) 1 (b) 3
(c) 3 and 4 (d) 1 and 4

25. In a mixture of 120 litres, the ratio of milk and water is 2 : 1. If the ratio of milk and water is 1 : 2, then the amount of water (in litres) is required to be added is :
(a) 20 (b) 40 (c) 80 (d) 120

26. A quantity x varies inversely as the square of y. Given that $x = 4$, when $y = 3$, the value of x when $y = 6$ is :
(a) 1 (b) 2 (c) 3 (d) 4

27. Suppose y varies as the sum of two quantities of which one varies directly as x and the other inversely as x. If $y = 6$ when $x = 4$ and $y = 3\frac{1}{3}$ when $x = 3$, then the relation between x and y is :
(a) $x = y + 4$ (b) $y = 2x + \frac{8}{x}$
(c) $y = 2x - \frac{8}{x}$ (d) $y = 2x - \frac{4}{x}$

28. The time period of a pendulum is proportional to the square root of the length of the pendulum. Consider the following statements :
(1) If the length of the pendulum is doubled, then the time period is also doubled.
(2) If the length is halved, then time period becomes one-fourth of the original time period.
The correct assertions are :
(a) 1 (b) 2
(c) neither 1 nor 2 (d) both 1 and 2

4.3 Unitary Method

In this method, at first we find the value of one unit and, then we find the value of required number of units by multiplying the value of one unit with the required number of units.

For example, if the price of 10 bananas is ₹ 50, find the price of 25 bananas. Then, as per unitary method, first we find the price of one unit of bananas, which is ₹ 5. Thus, we can find the price of 25 bananas by multiplying the price of one unit with the number of desired units, which is ₹ 125.

Direct Proportion

Two quantities are said to be directly proportional if the increase (or decrease) in one quantity causes the increase (or decrease) in the other quantity by same proportion. *e.g.,*

(i) The cost of articles varies directly with the number of articles. More articles more cost, less articles less cost.

(ii) The work done varies directly with the number of men (work force) at work. More men at work, more work done in the same time. Less men, less work done in the same time.

Inverse Proportion

Two quantities are said to vary inversely if the increase (or decrease) in one quantity causes the decrease (or increase) in the other quantity by same proportion. *e.g.,* The time taken to finish a work varies inversely to the number of men at work.

More men at work, less time taken to finish the same work. Less men at work, more time taken to finish the same work.

Exp. 1) If 6 note books cost ₹ 45, how much would 8 notebooks cost?

Solution More note books more cost; less note books, less cost

Note books	Cost
6	45
1	45/6
8	$\frac{8 \times 45}{6} = 60$

Hence 8 note books cost ₹ 60.

Exp. 2) If 45 students can consume a stock of food in 2 months, find for how many days the same stock of food will last for 27 students?

Solution More students, less days; less students, more days

Students	Days
45	60
1	60×45
27	$\frac{60 \times 45}{27} = 100$ days

Exp. 3) A man working 8 hours a day takes 5 days to complete a project. How many hours a day must he work to complete it in 4 days?

Solution 'More days, less hours; less days, more hours'

Days	Hours
5	8
1	5×8
4	$\frac{5 \times 8}{4} = 10$ hours a day

Introductory Exercise 4.2

1. If 40 articles cost ₹ 180, the cost of 18 articles is :
(a) ₹ 18 (b) ₹ 36
(c) ₹ 81 (d) ₹ 40.5

2. If 20 persons can do a piece of work in 7 days, then the number of persons required to complete the work in 28 days :
(a) 4 (b) 5
(c) 14 (d) 10

3. If 20 men can reap a field in 38 days, in how many days will 19 men reap the field?
(a) 21 days (b) 19 days
(c) 76 days (d) 40 days

4. A rope makes 260 rounds of a cylinder with base radius 20 cm. How many times can it go round a cylinder with base radius 26 cm?
(a) 130 (b) 300
(c) 200 (d) 150

5. A garrison of 750 men has provision for 20 weeks. If at the end of 4 weeks they are reinforced by 450 men, how long will the remaining provision last?
(a) 10 weeks (b) 11 weeks
(c) 15 weeks (d) 16 weeks

6. 40 men can build a wall 20 m high in 15 days. The number of men required to build a similar wall 25 m high in 6 days will be :
(a) 100 (b) 125
(c) 150 (d) 200

7. Cost of erecting a fence round a square field of 625 hectares at 15 paise per metre is :
(a) ₹ 500 (b) ₹ 750 (c) ₹ 100 (d) ₹ 1500

8. 56 workers can reap a field in 8 days. If the work is to be completed in 7 days, the extra workers needed are :
(a) 7 (b) 8 (c) 14 (d) 16

9. If 5 persons weave 180 shawls in 12 days, how many shawls will 6 persons weave in 15 days?
(a) 260 (b) 370 (c) 270 (d) 360

10. 15 men take 42 days of 4 hours each to do a piece of work. How many days of 6 hours each would 21 women take if 3 women do as much work as 2 men?
(a) 15 (b) 22 (c) 25 (d) 30

11. 10 engines consume 80 litres diesel when each is running 9 hours a day. How much diesel will be required for 10 engines, each running 15 hours a day, whereas 6 engines of the former type consume as much as 5 engines of latter type?
(a) 160 litre (b) 75 litre
(c) 80 litre (d) $111\frac{1}{9}$ litre

12. If $\frac{3}{5}$th of a cistern is filled in 30 minutes, how much more time will be required to fill the rest of it?
(a) 50 minutes (b) 20 minutes
(c) 15 minutes (d) 45 minutes

4.4 Variation

When two or more quantities are dependent upon each other and then if any one of them is changed, the other (dependent) quantity is also changed.

For example :

(i) When the salary of a person increases, then its savings/expenditure increases.

(ii) When the number of guests in a hotel/number of students in a hostel/number of employees changes, their respective expenses increases.

Basically, as it happens in direct proportion and inverse proportion, there are two types of variation :

(i) Direct variation (ii) Inverse variation

Direct Variation

A quantity A is said to vary directly if the increase (or decrease) in B yields increase (or decrease) in A but not in proportion. It is expressed as

$$A \propto B \quad \Rightarrow \quad A = KB,$$

where K is called proportionality constant

$$\Rightarrow \qquad K = \frac{A}{B}$$

Inverse Variation

A quantity A is said to vary inversely if the increase (or decrease) in B yields decrease (or increase) in A but not in same proportion. It is expressed as

$$A \propto \frac{1}{B} \quad \Rightarrow \quad A = \frac{K}{B}$$

or $K = AB$, K is called as proportionality constant.

NOTE

1. If it is not mentioned that a particular quantity is inversely variable, then it means the given quantity is directly variable.
2. A quantity sometimes vary jointly *i.e.,* directly on any quantity and inversely on another quantity.

e.g., $A \propto B$ and $A \propto \frac{1}{C}$

It means $A \propto \frac{B}{C} \Rightarrow A = \frac{KB}{C}$

Here A varies directly as B but inversely as C.

Also it can vary as only directly or inversely as more than one quantities.

e.g., $A \propto BC \Rightarrow A = KBC$

and $A \propto \frac{1}{BC} \Rightarrow A = \frac{K}{BC}$

Exp. 1) A varies directly as B and inversely as C. A is 12 when B is 6 and C is 2. What is the value of A when B is 12 and C is 3?

Solution $A \propto B$ and $A \propto \frac{1}{C}$

$\Rightarrow A \propto \frac{B}{C} \Rightarrow A = K\frac{B}{C}$

When $A = 12$, $B = 6$, $C = 2$, then

$12 = K\frac{6}{2} \Rightarrow K = 4$

Again $A = K\frac{B}{C} = 4 \times \frac{12}{3} = 16 \Rightarrow A = 16$

Exp. 2) The value of a coin varies directly to the square of its radius, when its thickness is constant. The radius of a coin is 1.5 cm and its value is ₹ 2. What will be the radius of a coin if its value is ₹ 5?

Solution $V \propto r^2 \Rightarrow V = Kr^2$

$2 = K \times (1.5)^2 \Rightarrow K = \frac{2}{2.25} \Rightarrow K = \frac{8}{9}$

Again $5 = \frac{8}{9} \times r^2 \Rightarrow r^2 = \frac{5 \times 9}{8}$

$\therefore r = \frac{3}{2} \times \sqrt{\frac{5}{2}} = 1.5 \times \sqrt{2.5} = 1.5 \times 1.6$

$r \approx 2.4$ cm

Hence, required radius = 2.4 cm

4.5 Problems Based on Ages

This article is very suitable as an appendix of ratio-proportion, since most of the questions based on ages involve the concept of ratio-proportion. *e.g.,* the age of Ravi is 16 years and the age of her mother is twice etc. or the ratio of the ages of the father and son at present is 3 : 1. 4 years, earlier the ratio was 4 : 1. What are the present ages of the father and son etc.

Exp. 1) The ratio of ages of Krishna and Balram is 3 : 4. Four years earlier the ratio was 5 : 7. Find the present ages of Krishna and Balram :

(a) 15 years, 20 years (b) 24 years, 32 years

(c) 16 years, 20 years (d) 32 years, 24 years

Solution Let the present age of Krishna and Balram be $3x$ and $4x$, then four years ago their ages be $(3x - 4)$ and $(4x - 4)$

So $\frac{(3x-4)}{(4x-4)} = \frac{5}{7} \Rightarrow 7(3x - 4) = 5(4x - 4)$

$\Rightarrow 21x - 28 = 20x - 20$

$\Rightarrow x = 8$

$\therefore$ Present age of Krishna = $3x = 24$ years

and age of Balram = $4x = 32$ years

Exp. 2) The ratio of age of Aman and her mother is 3 : 11. The difference of their ages is 24 years. What will be the ratio of their ages after 3 years?

Solution Let the present age of Aman and her mother be $3x$ and $11x$ then 3 years later their ages will be $(3x + 3)$ and $(11x + 3)$ respectively.

Again $11x - 3x = 24$

$8x = 24$

$\Rightarrow x = 3$

Therefore the ratio of their ages 3 years after

$= \frac{3x + 3}{11x + 3} = \frac{12}{36} = \frac{1}{3}$

Exp. 3) The age of Sachin is 4 times that of his son. Five years ago Sachin was nine times as old as his son was at that time. The present age of the Sachin is :

(a) 25 years (b) 36 years

(c) 32 years (d) 48 years

Solution Let the age of son is x years, then the age of Sachin will be $4x$ years.

$\therefore (4x - 5) = 9(x - 5) \Leftrightarrow x = 8$

$\therefore$ Age of Sachin is 32 years.

Exp. 4) The ratio of Varun's age and his mother's age is 5 : 11. The difference of their ages is 18 years. The ratio of their ages after 5 years will be :

(a) 19 : 59 (b) 2 : 3

(c) 37 : 75 (d) 10 : 19

Solution Let their ages be $5x$ and $11x$.

$11x - 5x = 18$

$\Leftrightarrow x = 3$

So their present ages are 15 and 33 years. Therefore, ratio of their ages after 5 years = 20 : 38 = 10 : 19.

Introductory Exercise 4.3

1. Amit is as much younger to Barkha as he is older to Chaman. If the sum of the ages of Barkha and Chaman is 48 years, what is the present age of Amit?
(a) 18 years
(b) 36 years
(c) 24 years
(d) 28 years

2. Bipin is 6 times old as Alok. Bipin's age will be twice of Chandan's age after 10 years. If Chandan's 7th birthday was celebrated 3 years ago, what is Alok's present age?
(a) 15 years
(b) 12 years
(c) 5 years
(d) none of these

3. Renuka got married 8 years ago. Today her age is $1\frac{1}{3}$ times her age at the time of marriage. Her daughter's age is 1/8 times her age. Her daughter's age is :
(a) 3 years (b) 4 years (c) 6 years (d) 8 years

4. Ten years ago *B* was twice of *A* in age. If the ratio of their present ages is 4 : 3, what is the sum of their present ages?
(a) 25 years (b) 30 years
(c) 40 years (d) 35 years

5. The sum of the ages of Aryabhatta and Shridhar is 45 years. Five years ago the product of their ages was 4 times the Aryabhatta's age at that time. The present ages of Aryabhatta and Shridhar respectively are :
(a) 25 and 20 (b) 35 and 10
(c) 36 and 9 (d) 40 and 5

4.6 **Partnership**

When two or more than two people run a business jointly by investing their money/resources, then it is called a joint venture or the business in partnership. All these people, who have invested their resources, are called as Partners.

Types of Partners

(i) **Working partner** A partner who is directly involved with day-to-day activities of business is called as working partner.

(ii) **Sleeping partner** A partner who just invests his or her money is called as sleeping partner.

General Rules of Partnership

(i) If the partners invest different amounts for the same period of time, then the profits of all the partners are shared in the ratio of their investments.

(ii) If the partners invest same amount for the different time periods, then the profits of all the partners are shared in the ratio of time periods for which their amounts were invested.

(iii) If the partners invest different amounts for different time periods, then their profits are shared in the ratio of products of respective investments with the time period for each partner, individually. Thus gain or loss is divided in the ratio of 'money-time' capitals.

NOTE Sometimes different problems are solved on the basis of partnership to find the expenses.

Exp. 1) Bhanu and Shafeeq started a business by investing ₹ 36000 and ₹ 63000. Find the share of each, out of an annual profit of ₹ 5500.

Solution Ratio of shares of Bhanu and Shafeeq
$= 36000 : 63,000 = 4 : 7$

$\therefore$ Share of Bhanu $= 5500 \times \frac{4}{11} = ₹\ 2000$

and Share of Shafeeq $= 5500 \times \frac{7}{11} = 3500$

Exp. 2) *A* starts some business with ₹ 50000. After 3 months *B* joins him with ₹ 70000. At the end of the year, in what ratio should they share the profits?

Solution Ratio of amount of *A* and *B* $= 50000 : 70000$

Ratio of time periods for *A* and *B* $= 12 : 9$

$\therefore$ Ratio of their money-time capital

Investments $= 50000 \times 12 : 70000 \times 9$
$= 20 : 21$

Exp. 3) Harsh Vardhan started a business by investing ₹ 36000. After 4 months Gyan Vardhan joined him with some investment. At the end of the year, the total profit was divided between them in the ratio of 9 : 7. How much capital was invested by Gyan Vardhan in the business?

Solution $\frac{36,000 \times 12}{x \times 8} = \frac{9}{7} \Rightarrow x = 42,000$

$\therefore$ Gyan Vardhan invested ₹ 42000 for 8 months only.

Exp. 4) A started some business with ₹ 26000. After 3 months B joined him with ₹ 16000. After some more time C joined them with ₹ 25000. At the end of the year, out of a total profit of ₹ 15453, C gets ₹ 3825 as his share. How many months after B joined the business did C join?

Solution Ratio (of share) of profits

$= 26,000 \times 12 : 16000 \times 9 : 25000 \times C$

$= 312 : 144 : 25C$

Now C's share $= \dfrac{25C}{456 + 25C} = \dfrac{3825}{15453} \Rightarrow C = 6$

Therefore C joined 3 months later than B joined.

Exp. 5) A, B and C started a business with their investments in the ratio 1 : 2 : 4. After 6 months A invested the half amount more as before and B invested twice the amount as before while C withdrew $\frac{1}{4}$th of the their investments. Find the ratio of their profits at the end of the year.

Solution Let us assume their initial investments were x, $2x$ and $4x$ respectively.

Therefore, ratio of their investments during the whole year

$= (x \times 6 + \dfrac{3x}{2} \times 6) : (2x \times 6 + 4x \times 6) : (4x \times 6 + 3x \times 6)$

$= 15x : 36x : 42x = 5x : 12x : 14x = 5 : 12 : 14$

$\therefore$ Ratio of their profits = 5 : 12 : 14

Exp. 6) A started a business with ₹ 52000 and after 4 months B joined him with ₹ 39000. At the end of the year, out of the total profits B received total ₹ 20000 including 25% of the profits as commission for managing the business. What amount did A receive?

Solution Profit's share of A and B

$= 52,000 \times 12 : 39,000 \times 8 = 2 : 1$

Let the profit be ₹ x, then B receives 25% as commission for managing business, the remaining 75% of the total profit x is shared between A and B in the ratio 2 : 1. Hence B will get $\frac{1}{3}$rd part of this in addition to his commission. Hence his

total earning $= 0.25x + \dfrac{1}{3} \times 0.75x = 0.5x = 20000 \Rightarrow x = 40000$

So, the remaining profit goes to A, hence the profit of A is ₹ 20000.

Exp. 7) A working partner gets 20% as his commission of the profit after his commission is paid. If the working partner's commission is ₹ 8000, then what is the total profit in the business?

Solution Let the total profit be ₹ x. The remaining profit after paying 20% working partner's commission = $(x - 8000)$. Again since 20% of this is working partner's commission,

therefore $\dfrac{20}{100} \times (x - 8000) = 8000 \Rightarrow x = 48000$

$\therefore$ The total profit in the business is ₹ 48000.

Introductory Exercise 4.4

1. A company make a profit of ₹ 9,00,000, 20% of which is paid as taxes. If the rest is divided among the partners P, Q and R in the ratio of $1 : 1\frac{1}{2} : 2$, then the shares of P, Q and R are respectively :
(a) 240000; 320000; 160000
(b) 320000; 240000; 160000
(c) 160000; 320000; 240000
(d) 160000; 240000; 320000

2. We have to divide a sum of ₹ 13950 among three persons A, B and C. B must get the double of A's share and C must get ₹ 50 less than the double of B's share. The share of A will be :
(a) ₹ 1950 (b) ₹ 1981.25
(c) ₹ 2000 (d) ₹ 2007.75

3. A started business with ₹ 45000 and B joined after- ward with ₹ 30000. If the profits at the end of one year were divided in the ratio 2 : 1 respectively, then B would have joined A for business after :
(a) 1 month (b) 2 months
(c) 3 months (d) 4 months

4. A and B are partners in a business. They invest in the ratio 5 : 6, at the end of 8 months A withdraws. If they receive profits in the ratio of 5 : 9, find how long B's investment was used?
(a) 12 months
(b) 10 months
(c) 15 months
(d) 14 months

5. Four milkmen rented a pasture. A put to graze 16 cows for 3 months, B 20 cows for 4 months, C 18 cows for 6 months and D 42 cows for 2 months. If A's share of rent be ₹ 2400, the rent paid by C is :
(a) ₹ 3200 (b) ₹ 4200 (c) ₹ 4000 (d) ₹ 5400

6. A, B and C subscribe ₹ 47000 for a business. If a subscribes ₹ 7000 more than B and B ₹ 5000 more than C, then out of total profit of ₹ 4700, C receives :
(a) ₹ 1200
(b) ₹ 4500
(c) ₹ 1000
(d) none of the above

CAT-Test

Questions Helping you bell the CAT

LEVEL 01 > BASIC LEVEL EXERCISE

1 Four numbers are in proportion. The sum of the squares of the four numbers is 50 and the sum of the means is 5. The ratio of first two terms is 1 : 3. What is the average of the four numbers?
(a) 2 (b) 3
(c) 5 (d) 6

2 A naughty student breaks the pencil in such a way that the ratio of two broken parts is same as that of the original length of the pencil to one of the larger part of the pencil. The ratio of the other part to the original length of pencil is
(a) $1 : 2\sqrt{5}$ (b) $2 : (3 + \sqrt{5})$
(c) $2 : \sqrt{5}$ (d) can't be determined

3 If $\frac{x+y}{x-y} = \frac{5}{3}$ and $\frac{x}{(y+2)} = 2$, then the value of (x, y) is
(a) (4, 1) (b) (2, 8)
(c) (1, 4) (d) (8, 2)

4 If $a^3 + b^3 : a^3 - b^3 = 133 : 117$; then find the value of $a : b$
(a) 2 : 3 (b) 5 : 4
(c) 5 : 2 (d) none of these

5 A student obtained equal marks in History and Sociology. The ratio of marks in Sociology and Geography is 2 : 3 and the ratio of marks in History and Philosophy is 1 : 2. If he has scored an aggregate of 55% marks. The maximum marks in each subject is same. In how many subjects did he score equal to or greater than 60% marks?
(a) 1 (b) 2
(c) 3 (d) none of these

6 The ratio of income of Anil and Mukesh is 2 : 3. The sum of their expenditure is ₹ 8000 and the amount of savings of Anil is equal to the amount of expenditure of Mukesh. What is the sum of their savings?
(a) 22000 (b) 4000
(c) 16000 (d) 12000

7 Hutch and Essar entered into a partnership just 5 months ago. The ratio of profit claimed by Hutch and Essar is 6 : 17. If Essar had just started his business 12 months ago with ₹ 1275, what is the amount contributed by Hutch?
(a) ₹ 980 (b) ₹ 1080
(c) ₹ 1200 (d) ₹ 998

8 A child has three different kinds of chocolates costing ₹ 2, ₹ 5 and ₹ 10. He spends total ₹ 120 on the chocolates. What is the minimum possible number of chocolates, he can buy, if there must be atleast one chocolate of each kind?
(a) 22 (b) 19 (c) 17 (d) 15

9 In the previous problem (no. 8) what is the maximum possible no. of chocolates?
(a) 52 (b) 53 (c) 55 (d) 60

10 Mr. Teremere and Mr. Meretere have 5 chocolates and 3 chocolates with them respectively. Meanwhile Mr. Khabbu Singh joined them and all 8 chocolates were distributed equally among all these three people. In turn Khabbu Singh gave ₹ 16 to Mr. Teremere and Mr. Meretere, since Khabbu Singh did not has any chocolate. What is the difference of amounts received by Teremere and Meretere? Given that the amount was shared in proportion of chocolates received by Khabbu Singh.
(a) ₹ 8 (b) ₹ 12 (c) ₹ 14 (d) ₹ 15

11 ₹ 4536 is divided among 4 men, 5 women and 2 boys. The ratio of share of a man, a woman and a boy is 7 : 4 : 3. What is the share of a woman?
(a) ₹ 336 (b) ₹ 498 (c) ₹ 166 (d) ₹ 256

12 The concentration of petrol in three different mixtures (petrol and kerosene) is $\frac{1}{2}$, $\frac{3}{5}$ and $\frac{4}{5}$ respectively. If 2 litres, 3 litres and 1 litre are taken from these three different vessels and mixed. What is the ratio of petrol and Kerosene in the new mixture?
(a) 4 : 5 (b) 3 : 2 (c) 3 : 5 (d) 2 : 3

13 Time period (T) of pendulum is directly proportional to the square root of length of string by which bob is attached to a fixed point and inversely proportional to the square root of gravitational constant 'g'. Time period of a bob is 3 seconds when the gravitational constant g is 4 m/sec^2 and length of string is 9 metre, what is the time period of a bob having a string of length 64 metre and gravitational constant 16 m/sec^2?
(a) 4 seconds (b) 12 seconds
(c) 16 seconds (d) 10 seconds

14 In a milk shoppe there are three varieties of milk, 'Pure', 'Cure' and 'Lure'. The 'Pure' milk has 100% concentration of milk. The ratio of milk is to water in the 'Cure' is 2 : 5 and in the Lure it is 3 : 8 respectively. Sonali purchased 14 litres of Cure and 22 litres of Lure milk and mixed them. If she wanted to make the concentration of milk in the mixture of purchased milk to 50%. How many litres of 'Pure' milk she is needed?
(a) 6 litres (b) 8 litres
(c) 16 litres (d) 18 litres

15 In the squadron of Indian Air Force the ratio of Sukhoi is to Mig and Jaguar together is 5 : 7 and the ratio of Jaguar is to Sukhoi and Mig together is 1 : 2. Find the ratio of Sukhoi and Mig :
(a) 2 : 7 (b) 3 : 5 (c) 3 : 1 (d) 5 : 3

16 During our campaign against child labour we have found that in three glass making factories A, B and C there were total 33 children aged below 18 were involved. The ratio of male to female in A, B and C was 4 : 3, 3 : 2 and 5 : 4 respectively. If the no. of female children working in the factories B and C be equal then find the no. of female children working in factory A :
(a) 5 (b) 2 (c) 8 (d) 6

17 The value of a diamond is directly proportional to the square of its weight. A diamond unfortunately breaks into three pieces with weights in the ratio of 3 : 4 : 5 thus a loss of ₹ 9.4 lakh is incurred. What is the actual value of diamond :
(a) 28.8 lakh (b) 13.5 lakh
(c) 14.4 lakh (d) 18.8 lakh

18 In the Ruchika's wallet there are only ₹ 16, consisting of 10 paise, 20 paise and ₹ 1 coins. The ratio of no. of coins of 10 paise and 20 paise is 6 : 1. The minimum no. of ₹ 1 coin is :
(a) 5 (b) 12 (c) 4 (d) 8

19 There are two vessels containing the mixture of milk and water. In the first vessel the water is 2/3 of the milk and in the second vessel water is just 40% of the milk. In what ratio these are required to mix to make 24 litres mixture in which the ratio of water is to milk is 1 : 2 ?
(a) 4 : 3 (b) 5 : 7
(c) 5 : 2 (d) 7 : 5

20 Nehru Ji had 'n' chocolates. He distributed them among 4 children in the ratio of $\frac{1}{2} : \frac{1}{3} : \frac{1}{5} : \frac{1}{8}$. If he gave them each one a complete chocolate, the minimum no. of chocolates that he had :
(a) 139 (b) 240
(c) 278 (d) None of these

21 The ratio of working efficiency of A and B is 5 : 3 and the ratio of efficiency of B and C is 5 : 8. Who is the most efficient ?
(a) A (b) B
(c) C (d) Can't be determined

22 If $4A = 5B$ and $3A = 2C$, the ratio of $B : C$ is :
(a) 4 : 3 (b) 5 : 8 (c) 8 : 15 (d) 10 : 15

23 Equal quantities of three mixtures of milk and water are mixed in the ratio of 1 : 2, 2 : 3 and 3 : 4. The ratio of water and milk in the mixture is :
(a) 193 : 122 (b) 122 : 193
(c) 61 : 97 (d) 137 : 178

24 The ratio of age of A and B is 8 : 9 and the age of B is 2/3 of C's age and age of C is $\frac{9}{13}$ times the age of D. If the age of B is 18 years then the age of C is :
(a) 36 years (b) 39 years
(c) 27 years (d) 54 years

25 A milk man has a mixture of milk in which ratio of milk and water is 5 : 3. He sells 40 litres of milk *i. e.*, mixture then he adds up 15 litres of pure water. Now the ratio of milk and water is 5 : 4. What is the new quantity of mixture?
(a) 72 litres (b) 270 litres
(c) 135 litres (d) Data insufficient

26 A and B are two alloys of copper and tin prepared by mixing the respective metals in the ratio of 5 : 3 and 5 : 11 respectively. If the alloys A and B are mixed to form a third alloy C with an equal proportion of copper and tin, what is the ratio of alloys A and B in the new alloy C?
(a) 3 : 5 (b) 4 : 5 (c) 3 : 2 (d) 2 : 3

27 A hotel incurs two types of expenses, one which is fixed and others depend upon no. of guests. When there are 10 guests, total expenses of hotel are ₹ 6000. Also when there are 25 guests average expenses per guests are ₹ 360? What is the total expenses of hotel when there are 40 guests?
(a) ₹ 8000 (b) ₹ 12000
(c) ₹ 15500 (d) none of these

28 The ratio of third proportional to 21 and 42 and mean proportional to 16 and 49 is :
(a) 3 : 1 (b) 2 : 3 (c) 4 : 3 (d) 1 : 3

29 The period of the pendulum is directly proportional to the square root of the length of the string. The period of such a pendulum with string of length 16 cm is 52 seconds. Find the length of the string if the period is 65 seconds :
(a) 4.5 cm (b) 5 cm
(c) 6 cm (d) none of these

30 For any two numbers m, n;$(m + n) : (m - n) : mn = 7 : 1 : 60$, then find the value of $\frac{1}{m} : \frac{1}{n}$
(a) 4 : 3 (b) 8 : 6 (c) 3 : 4 (d) 7 : 8

31 ₹ 960 were distributed among A, B, C and D in such a way that C and D together gets half of what A and B together gets and C gets one-third amount of B. Also D gets $\frac{5}{3}$ times as much as C. What is the amount of A?
(a) ₹ 240 (b) ₹ 280
(c) ₹ 320 (d) data insufficient

32 Find the value of $\frac{p^2+q^2}{r^2+s^2}$, if $p:q::r:s$.

(a) $\frac{1}{4}$ (b) $\frac{1}{9}$ (c) $\frac{ps}{rq}$ (d) $\left(\frac{p-q}{r-s}\right)^2$

33 Find the value of $\frac{p^2+q^2}{r^2+s^2}$, if $p:q::r:s$.

(a) $\frac{1}{4}$ (b) $\frac{pq}{rs}$

(c) $\left(\frac{p+q}{r+s}\right)^2$ (d) Both (b) and (c)

34 The speeds of rickshaw, car and scooter are in the ratio of 3 : 5 : 6. What is the ratio of time taken by each one of them for the same distance?

(a) 6 : 5 : 3
(b) 10 : 6 : 5
(c) 12 : 7 : 6
(d) Data insufficient

35 Divide ₹ 6940 in such a way that A gets $\frac{2}{3}$rd of what B gets and B gets $\frac{3}{5}$th of what C gets? What is the share of A and B together?

(a) ₹ 1982 (b) ₹ 1388
(c) ₹ 3470 (d) none of these

36 The ratio of age of A and B is $x:y$. If A's age is increased by 3 years and B's age is increased by 2 years then new ratio of their ages becomes 24 : 25. Given that the sum of their actual ages is 93 years. Find the actual ratio of their ages.

(a) 21 : 22 (b) 42 : 45
(c) 45 : 48 (d) Can't be determined

37 $a:b=4:9$ if 4 is added to both of the numbers then the new ratio becomes 21 : 46. What is the difference between a and b?

(a) 80 (b) 100
(c) 125 (d) 130

38 The ratio of ages of Rahul and Deepesh is 3 : 5. 10 years later this ratio becomes 5 : 7. What is the present age of Deepesh?

(a) 20 years (b) 50 years
(c) 25 years (d) 40 years

39 When 5 is added to the numerator and denominator both of a (positive) fraction, then the new ratio of numerator to denominator becomes 11 : 15. What is the original ratio?

(a) 17 : 25 (b) 3 : 5
(c) 28 : 40 (d) None of these

40 Five numbers a, b, c, d and e are in the ratio of 2 : 3 : 5 : 8 : 9 and their sum is 162. Find the average of all these numbers

(a) 27 (b) 30
(c) 32.4 (d) Can't be determined

41 6 pumps of Kirlosker can fill a tank in 7 days and 2 similar pumps of USHA can fill the same tank in 18 days. What is the ratio of the efficiency of a Kirlosker pump and a USHA pump?

(a) 6 : 7 (b) 7 : 6
(c) 7 : 54 (d) can't be determined

42 16 persons can reap $\frac{1}{5}$th field in 6 days. How many persons (with same efficiency) are required to reap rest of the field in 8 days?

(a) 27 (b) 54
(c) 48 (d) 64

43 The LCM of two numbers is 210 and their ratio is 2 : 3. The sum of these numbers is :

(a) 210 (b) 175
(c) 315 (d) can't be determined

44 What number must be subtracted from each of the numbers 53, 21, 41, 17 so that the remainders are in proportion?

(a) 1 (b) 3
(c) 5 (d) none of these

45 The angles of a triangle are in the ratio of 2 : 3 : 4. Find the measurement of greatest angle :

(a) 30° (b) 60°
(c) 100° (d) 80°

46 In a wallet the ratio of 25 paise, 50 paise and ₹ 1 coins are in the ratio of 12 : 4 : 3, which amounts to ₹ 600. Find the no. of coins of 25 paise :

(a) 200 (b) 225
(c) 275 (d) none of these

47 $(x-a):(x-b):(x-c)=11:9:5$, where $x=\frac{a+b+c}{2}$.

What is the ratio of a, b, c ?

(a) 10 : 8 : 7 (b) 3 : 5 : 9 (c) 7 : 8 : 10 (d) 6 : 5 : 3

48 Petrol is 7 times heavy than Kerosene and Castrol mobil is 18 times as heavy as Kerosene. What should be the ratio of petrol and mobil in the new mixture to get the mixture which must be 11 times as heavy as kerosene?

(a) 3 : 4 (b) 7 : 4
(c) 9 : 19 (d) 9 : 10

49 A girl buys 2 pigeons for ₹ 182. She sells one at a loss of 5% and another at a profit if 8%. But she neither gains nor loses on the whole. Find the price of pigeon which has sold at a profit :

(a) ₹ 112 (b) ₹ 85
(c) ₹ 70 (d) can't be determined

50 The ratio of prices of Cello and Rotomac pens in 2000 were in the ratio of 3 : 5. In 2005 the price of Cello pen trebles itself and the price of Rotomac pen is increased by ₹ 100, then the new ratio of prices of the same pens becomes 4 : 5. What was the original price of the Rotomac pen in 2000?

(a) ₹ 60 (b) ₹ 80
(c) ₹ 100 (d) ₹ 120

51 A goldsmith has 361 rings of gold. He sells some of them at a loss of 4% and rest at a profit of 15% making overall profit of 8%. Find the no. of rings sold at a profit of 15% :
(a) 171 (b) 133
(c) 218 (d) 228

52 Michel travelled from New York to New Jersey covering total distance of 250 mile in 8 h partly by car at 30 mile/h and rest by train at 35 mile/h. The distance travelled by car is :
(a) 150 mile (b) 80 mile
(c) 220 mile (d) 180 mile

53 A rabit takes 22 leaps for every 17 leaps of cat and 22 leaps of a rabit are equal to 17 leaps of the cat. What is the ratio of the speeds of rabit and cat?
(a) 1 : 1 (b) 484 : 289
(c) 17 : 22 (d) none of these

54 ₹ 171 are divided among four friends in the ratio of $\frac{1}{3}:\frac{1}{4}:\frac{1}{5}:\frac{1}{6}$. What is the amount of the person who got the greatest share?
(a) 14 (b) 40
(c) 36 (d) 60

55 10 years ago the age of Karishma was $\frac{1}{3}$rd of the age of Babita. 14 years hence the ratio of ages of Karishma and Babita will be 5 : 9. Find the ratio of their present ages :
(a) 13 : 29 (b) 11 : 27
(c) 29 : 17 (d) 13 : 25

56 The ratio of numerator to a denominator of a fraction is $\frac{1}{5}$ when x and $5x$ are added to the numerator and denominator respectively to the given fraction then the ratio of the new fraction will be :
(a) 1 : 1 (b) 1 : 25 (c) 1 : 5 (d) 2 : 7

57 x varies directly as y and x varies inversely as the square of z. When $y = 75$ and $x = 6$, then $z = 5$. Find the value of x when $y = 24$ and $z = 4$:
(a) 1 (b) 2 (c) 3 (d) 4

58 x varies directly as $(y^2 + z^2)$. At $y = 1$ and $z = 2$, the value of x is 15. Find the value of z, when $x = 39$ and $y = 2$:
(a) 2 (b) 3 (c) 4 (d) 6

59 Weight of a sumo is jointly varies as his height and his age. When height is 1.2 m and age is 20 years his weight is 48 kg. Find the weight of the sumo when his height is 1.5 metre and age is 30 years :
(a) 60 kg (b) 72 kg (c) 90 kg (d) 58 kg

60 If $(a + b) : (b + c) : (c + a) = 5 : 6 : 9$ and $a + b + c = 10$. What is the value of c :
(a) 2 (b) 3 (c) 5 (d) 7

61 A, B and C have amounts in the ratio of 3 : 4 : 5. First B gives $\frac{1}{4}$th to A and $\frac{1}{4}$th to C then C gives $\frac{1}{6}$th to A. Find the final ratio of amount of A, B and C, respectively :
(a) 4 : 3 : 5 (b) 5 : 4 : 3
(c) 6 : 4 : 2 (d) 5 : 2 : 5

LEVEL 02 > HIGHER LEVEL EXERCISE

1 A tin contains a mixture of Dew and Sprite in the ratio of 7 : 3 and another tin contains the Dew and Sprite in the ratio of 5 : 4. In what proportion should the solution of two tins be mixed to achieve a perfect proportion of 2 : 1 (in which Dew is 2 times that of sprite).
(a) 10 : 3 (b) 4 : 1
(c) 3 : 10 (d) 3 : 1

2 There are two containers A and B filled with mentha oil with different prices and their volumes are 140 litres and 60 litres respectively. Equal quantities are drawn from both A and B in such a manner that the oil drawn from A is poured into B and the oil drawn from B is poured into A. The price per litre becomes equal in both A and B. How much oil is drawn from each of A and B :
(a) 40 litre (b) 21 litre
(c) 42 litre (d) can't be determined

3 A three digit number is such that this number itself is divisible by the sum of its digits. The sum of hundreds and unit digit is 6 while the sum of the tens and unit digit is 5. What is the ratio of unit and tens digit :
(a) 1 : 2 (b) 2 : 3 (c) 3 : 4 (d) 2 : 7

4 Sachin bought 1.5 kg fresh grapes. The ratio of water is to pulp was 4 : 1. When his naughty child crushed these grapes, then some water get wasted. Now the ratio of water is to pulp is 3 : 2. What is the total amount of the crushed grapes?
(a) 0.5 kg (b) 1 kg
(c) 0.75 kg (d) none of these

5 A man bought 9 mangoes for a rupee and sold them at 6 mangoes for a rupee. What is the ratio of profit to the cost price?
(a) $\frac{3}{10}$ (b) $\frac{3}{2}$
(c) 1/2 (d) none of these

6 Half of the volume of Petrol and kerosene mixture of ratio 7 : 5 is converted into a mixture of ratio 3 : 1 by the substitution (or replacement) method. While the mixture of ratio 7 : 5 was formed from the mixture of 7 : 3 by adding the Kerosene in it. If 240 litres petrol is required in the replacement method, what is the total amount of Kerosene was added to prepare the mixture of 7 : 5 ?
(a) 100 litres (b) 400 litres
(c) 50 litres (d) 200 litres

11 Dia and Urea are two chemical fertilizers. Dia is consists of N, P and K and Urea conists of only N and P. A mixture of Dia and Urea is prepared in which the ratio of N, P and K is 26%, 68% and 6% respectively. The ratio of N, P and K in Dia is 20%, 70% and 10% respectively. What is the ratio of N and P in the Urea?
(a) 27% and 63%
(b) 33% and 67%
(c) 35% and 65%
(d) 70% and 30%

12 The ratio of copper and nickel by weight in the two alloys X and Y are 2 : 7 and 5 : 4. How many kilogram of the alloy X and Y are required to make 42 kg of new alloy Z in which the ratio of copper and nickel is same?
(a) 6 kg and 36 kg (b) 10 kg and 32 kg
(c) 7 kg and 35 kg (d) none of these

13 There are two alloys made up of copper and aluminium. In the first alloy copper is half of the aluminium and in the second alloy copper is thrice as much as aluminium. How many times the second alloy must be mixed with first alloy to get the new alloy in which copper is twice as that of aluminium?
(a) 2 (b) 3
(c) 4 (d) 5

14 There are 90 litres castrol and 150 litres CRB mobil oils. The price of castrol is ₹ 80 per litre and price of CRB is ₹ 75 per litre. Equal amount of castrol and CRB is taken out and then CRB is poured out in the vessel of castrol and castrol is poured out in the vessel of CRB. Now the rate of both the mixtures is same. What is the amount of mobil oil taken out from each of the vessel?
(a) 45 litres (b) 56.25 litres (c) 24.5 litres (d) 36 litres

15 There are two containers, the first contain, 1 litre pure water and the second contains 1 litre of pure milk. Now 5 cups of water from the first container is taken out is mixed well in the second container. Then, 5 cups of this mixture is taken out and is mixed in the first container. Let A denote the proportion of milk in the first container and B denote the proportion of water in the second container then :
(a) $A < B$ (b) $A = B$
(c) $A > B$ (d) can't be determined

16 Mitthu Bhai sells rasgulla (a favourite indian sweets) at ₹ 15 per kg. A rasgulla is madeup of flour and sugar in the ratio of 5 : 3. The ratio of price of sugar and flour is 7 : 3 (per kg). Thus he earns $66\frac{2}{3}$% profit. What is the cost price of sugar?
(a) ₹ 10/kg (b) ₹ 9/kg (c) ₹18/kg (d) ₹14/kg

Answers

✓ *Introductory Exercise 4.1*

1 *(b)*	2 *(d)*	3 *(c)*	4 *(a)*	5 *(d)*	6 *(d)*	7 *(c)*	8 *(b)*	9 *(a)*	10 *(d)*
11 *(b)*	12 *(d)*	13 *(b)*	14 *(c)*	15 *(c)*	16 *(b)*	17 *(c)*	18 *(d)*	19 *(b)*	20 *(b)*
21 *(c)*	22 *(a)*	23 *(c)*	24 *(b)*	25 *(d)*	26 *(a)*	27 *(c)*	28 *(c)*		

✓ *Introductory Exercise 4.2*

1 *(c)*	2 *(b)*	3 *(d)*	4 *(c)*	5 *(a)*	6 *(b)*	7 *(d)*	8 *(b)*	9 *(c)*	10 *(d)*
11 *(a)*	12 *(b)*								

✓ *Introductory Exercise 4.3*

1 *(c)*	2 *(c)*	3 *(b)*	4 *(d)*	5 *(c)*

✓ *Introductory Exercise 4.4*

1 *(d)*	2 *(c)*	3 *(c)*	4 *(a)*	5 *(d)*	6 *(c)*

✓ **Level 01** *Basic Level Exercise*

1 *(b)*	2 *(b)*	3 *(d)*	4 *(c)*	5 *(b)*	6 *(d)*	7 *(b)*	8 *(c)*	9 *(b)*	10 *(b)*
11 *(a)*	12 *(b)*	13 *(a)*	14 *(c)*	15 *(d)*	16 *(d)*	17 *(c)*	18 *(c)*	19 *(b)*	20 *(a)*
21 *(a)*	22 *(c)*	23 *(a)*	24 *(c)*	25 *(c)*	26 *(c)*	27 *(b)*	28 *(a)*	29 *(d)*	30 *(c)*
31 *(b)*	32 *(d)*	33 *(d)*	34 *(b)*	35 *(c)*	36 *(c)*	37 *(b)*	38 *(c)*	39 *(d)*	40 *(c)*
41 *(a)*	42 *(c)*	43 *(b)*	44 *(c)*	45 *(d)*	46 *(d)*	47 *(c)*	48 *(b)*	49 *(c)*	50 *(b)*
51 *(d)*	52 *(d)*	53 *(a)*	54 *(d)*	55 *(a)*	56 *(c)*	57 *(c)*	58 *(b)*	59 *(c)*	60 *(c)*
61 *(d)*									

✓ Level 02 *Higher Level Exercise*

1 *(a)*	2 *(c)*	3 *(b)*	4 *(c)*	5 *(c)*	6 *(d)*	7 *(d)*	8 *(a)*	9 *(c)*	10 *(c)*
11 *(d)*	12 *(b)*	13 *(a)*	14 *(b)*	15 *(d)*	16 *(c)*	17 *(d)*	18 *(a)*	19 *(a)*	20 *(c)*
21 *(c)*	22 *(c)*	23 *(a)*	24 *(b)*	25 *(a)*	26 *(c)*	27 *(c)*	28 *(b)*	29 *(a)*	30 *(c)*
31 *(d)*	32 *(a)*	33 *(d)*	34 *(b)*	35 *(b)*	36 *(b)*	37 *(c)*	38 *(b)*	39 *(c)*	40 *(b)*
41 *(b)*	42 *(c)*	43 *(b)*	44 *(b)*	45 *(b)*	46 *(b)*	47 *(c)*	48 *(a)*	49 *(a)*	

✓ Level 03 *Final Round*

1 *(d)*	2 *(b)*	3 *(c)*	4 *(a)*	5 *(b)*	6 *(b)*	7 *(d)*	8 *(b)*	9 *(d)*	10 *(b)*
11 *(c)*	12 *(c)*	13 *(c)*	14 *(b)*	15 *(b)*	16 *(d)*				

Hints & Solutions

☑ Level 01 *Basic Level Exercise*

1 $a : b :: c : d$

$a^2 + b^2 + c^2 + d^2 = 50$

$b + c = 5$

and $a : b = 1 : 3$

If consider $a : b = 1 : 3$ as it is, then

$c = 2$ $(5 - 3 = 2)$

and $d = 6$ $(\because a : b :: c : d)$

$a^2 + b^2 + c^2 + d^2 = 1^2 + 3^2 + 2^2 + 6^2 = 50$

Hence, the presumed values are correct.

Thus, the average of a, b, c and $d = \dfrac{a + b + c + d}{4}$

$= \dfrac{1 + 3 + 2 + 6}{4} = 3$

Hence (b) is correct.

Alternatively Assume option (b)

$\because \quad \dfrac{a + b + c + d}{4} = 3$

$\Rightarrow \quad a + b + c + d = 12$

Now $\because \quad b + c = 5$

$\Rightarrow \quad a + d = 7$

Again $a : b = 1 : 3$

$\Rightarrow \quad c : d = 2 : 6$

Now verify that $a^2 + b^2 + c^2 + d^2 = 50$. Since it is correct.

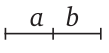
Hence option (b) is correct.

NOTE It can also be solved by forming quadratic equations.

2 $\dfrac{a}{b} = \dfrac{a + b}{a}$

$\Rightarrow \quad a^2 = ab + b^2$

$\Rightarrow \quad a^2 - b^2 - ab = 0$



Let $b = 1$, then $a : b = a : 1$ (by putting $b = 1$)

$\therefore \quad a^2 - a - 1 = 0$

$\Rightarrow \quad a = \dfrac{1 \pm \sqrt{5}}{2}$ (Solving quadratic equation by Sridharacharya's formula)

$\therefore \quad a = \dfrac{1 + \sqrt{5}}{2}$ (negative value can't be considered)

$a : b = \dfrac{1 + \sqrt{5}}{2} : 1$

or $a : b = (1 + \sqrt{5}) : 2$

Therefore, $\dfrac{b}{a + b} = \dfrac{2}{1 + \sqrt{5} + 2} = \dfrac{2}{3 + \sqrt{5}}$

Hence option (b) is correct.

3 The best way is to go through options

$\dfrac{x + y}{x - y} = \dfrac{8 + 2}{8 - 2} = \dfrac{10}{6} = \dfrac{5}{3}$ (verified)

Again $\dfrac{x}{y + 2} = \dfrac{8}{2 + 2} = 2$ (verified)

Hence option (d) is correct.

4 Best way is to go through options

and $\dfrac{a^3 + b^3}{a^3 - b^3} = \dfrac{5^3 + 2^3}{5^3 - 2^3} = \dfrac{125 + 8}{125 - 8} = \dfrac{133}{117}$

Hence option (c) is correct.

5 $H : S = 1 : 1$

and $S : G = 2 : 3$

$H : P = 1 : 2$

$\therefore \quad H : S : G : P = 2 : 2 : 3 : 4$

$= 2x : 2x : 3x : 4x$

Therefore $\dfrac{2x + 2x + 3x + 4x}{4} = \dfrac{11x}{4} = 55$

$\Rightarrow \quad x = 20$

Marks in History = 40
Sociology = 40
Geography = 60
Philosophy = 80

Hence, only in two subjects he scored 60% or above.
Hence option (b) is correct.

6 Let the incomes of A and M is $2x$ and $3x$.
Let the savings of A be K, then the expenditure of M be K
Also expenditure of $A = 2x - K$
Given $(2x - K) + K = 8000 \Rightarrow x = 4000$
$\therefore$ Total income of
A and $B = 2x + 3x = 5x = 5 \times 4000 = 20000$
$\therefore$ Total savings of A and $B = 20000 - 8000 = ₹\ 12{,}000$

7 $$\frac{\text{Profit of Hutch}}{\text{Profit of Essar}} = \frac{\text{time period} \times \text{amount of Hutch invested}}{\text{time period} \times \text{amount of Essar invested}}$$

$$\frac{6}{17} = \frac{5 \times K}{12 \times 1275}$$

$$\Rightarrow \quad K = \frac{6 \times 12 \times 1275}{17 \times 5} = 1080$$

8 Minimum number of chocolates are possible when he purchases maximum number of costliest chocolates.
Thus $2 \times 5 + 5 \times 2 = ₹\ 20$
Now ₹ 100 must be spend on 10 chocolates as $100 = 10 \times 10$.
Thus minimum number of chocolates $= 5 + 2 + 10 = 17$

9 Maximum number of chocolates are possible only when he purchases minimum number of costlier chocolates and maximum number of cheaper chocolates
$\therefore \quad 2 \times 5 + 1 \times 10 = ₹\ 20$
Now ₹ 100 must be spend on 50 chocolates as $100 = 2 \times 50$.
Thus maximum number of possible chocolates
$= 2 + 1 + 50 = 53$

10 Meretere $\rightarrow$ M, Teremere $\rightarrow$ T and Khabbu Singh $\rightarrow$ K
$\frac{5+3}{3} = 2.66$ (since all the 8 chocolates were shared by 3)
It means M has given $5 - 2.66 = 2.33$ chocolates to K
and T has given $3 - 2.66 = 0.33$ chocolates to K
Thus M and T will receive the amount in the ratio of donations (*i.e.,* share of chocolates)

M	:	T
2.33	:	0.33
$2\frac{1}{3}$	:	$\frac{1}{3}$
$\frac{7}{3}$	:	$\frac{1}{3}$
7	:	1

So the M receives ₹ 14 and T receives ₹ 2
Thus the difference = ₹ 12

11 Share of a man, a woman and a boy = $7x$, $4x$ and $3x$
then the share of 4 men = $4 \times 7x = 28x$
then the share of 5 women = $5 \times 4x = 20x$
then the share of 2 boys = $2 \times 3x = 6x$
Now, the share of all women $= \frac{20x}{(28x + 20x + 6x)} \times 4536$
$= \frac{20}{54} \times 4536 = ₹\ 1680$
Hence, the share of one woman $= \frac{1680}{5} = 336$

12 Concentration of petrol in

A	B	C
$\frac{1}{2}$	$\frac{3}{5}$	$\frac{4}{5}$

Quantity of petrol taken from $A = 1$ litre out of 2 litre
Quantity of petrol taken from $B = 1.8$ litre out of 3 litre
Quantity of petrol taken from $C = 0.8$ litre out of 1 litre
Therefore total petrol taken out from
A, B and $C = 1 + 1.8 + 0.8 = 3.6$ litre
So, the quantity of Kerosene $= (2 + 3 + 1) - (3.6) = 2.4$ litre
Thus, the ratio of petrol to Kerosene $= \frac{3.6}{2.4} = \frac{3}{2}$

13 $$T \propto \frac{\sqrt{l}}{\sqrt{g}} \Rightarrow T = k\sqrt{\frac{l}{g}}$$
Therefore, $3 = K\sqrt{\frac{9}{4}}$ or $3 = K\frac{3}{2} \Rightarrow K = 2$
Again $T = K\sqrt{\frac{l}{g}} = 2 \times \sqrt{\frac{64}{16}} = 4$
$T = 4$ seconds

14

Pure	Cure	Lure
100%	40%	37.5%
$\frac{1}{1}$	$\frac{2}{5}$	$\frac{3}{8}$

	Milk	Water	Mixture
Cure	$4l$	$10l$	$14l$
Lure	$6l$	$16l$	$22l$
New mixture	Milk	Water	
	$10l$	$26l$	
	$16l$		
	$26l$	$26l$	

Required mixture 1 : 1
Since in the required mixture the ratio of milk and water is 1 : 1 so she has to add up 16 litre of more milk (pure) to get it, for the fixed quantity of water.

15 $S : (M + J) = 5 : 7 \Rightarrow 7S = 5M + 5J$ ...(i)
$J : (S + M) = 1 : 2 \Rightarrow 2J = S + M$ ...(ii)
By solving eqs. (i) and (ii), we get
$S : M : J = 5 : 3 : 4$
So $S : M = 5 : 3$

16

	Male		Female
A	$4x$	:	$3x$
B	$3y$	:	$2y$
C	$5z$	:	$4z$

but $2y = 4z$

$\therefore$

	Male		Female
A	$4x$	:	$3x$
B	$6z$	:	$4z$
C	$5z$	:	$4z$

Therefore, $7x + 19z = 33$

but z can assume only one value *i.e.*, $z = 1$

Hence, $7x + 19 = 33 \Rightarrow x = 2$

Thus the no. of female children in factory $A = 3x = 6$.

17 The ratio of broken parts (by weight) $= 3x : 4x : 5x$

Therefore value of broken parts of diamond

$$= (3x)^2 + (4x)^2 + (5x)^2 = 50x^2$$

The value of original diamond $= (3x + 4x + 5x)^2 = 144x^2$

Therefore, loss in value $= 144x^2 - 50x^2 = 9.4$ lakh

$\Rightarrow 94x^2 = 9.4$ lakh

$\Rightarrow 94x^2 = 940000$

$\Rightarrow x^2 = 10000$

Hence, the actual value of the diamond $= 144x^2$

$= 144 \times 10000$

$= 14.4$ lakh

18. $10x + 20y + 100z = 1600$

Again since $x : y = 6 : 1$

$\therefore 60y + 20y + 100z = 1600$

$\Rightarrow 80y + 100z = 1600$

$\Rightarrow 4y + 5z = 80$

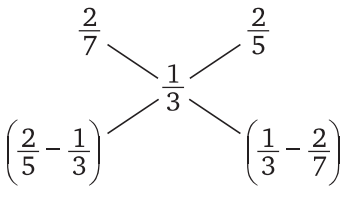
Putting $z = 1, 2, 3, 4, 5 \ldots$, we get at $z = 4$, $y = 15$ (an integer)

Hence, min. 4 coins of ₹ 1 will be there.

19 Concentration of water in first vessel $= \frac{2}{5} = 40\%$

Concentration of water in second vessel $= \frac{2}{7} = 28.57\%$

By alligation



$$\frac{1}{15} : \frac{1}{21}$$

$$21 : 15$$

$$7 : 5$$

Therefore, ratio of first mixture to the second mixture $= 5 : 7$

20 LCM of 2, 3, 5, 8 $= 120$

Therefore, $\frac{1}{2} \times 120 = 60$

$\frac{1}{3} \times 120 = 40$

$\frac{1}{5} \times 120 = 24$

$\frac{1}{8} \times 120 = 15$

Therefore, minimum number of chocolates

$= 60 + 40 + 24 + 15 = 139$

21 $A : B = 5 : 3$

$B : C = 5 : 8$

$\therefore A : B : C = 25 : 15 : 24$

So, A is the most efficient.

22 $A : B = 5 : 4 \Rightarrow 10 : 8$

$A : C = 2 : 3 \Rightarrow 10 : 15$

$\therefore A : B : C = 10 : 8 : 15$

$\therefore B : C = 8 : 15$

23

(Proportion of milk in mixture)	A		B		C
	$\frac{1}{3}$	:	$\frac{2}{5}$	:	$\frac{3}{7}$
or	$\frac{35}{105}$	:	$\frac{42}{105}$	:	$\frac{45}{105}$

Quantity of milk in new mixture $= 35 + 42 + 45 = 122l$

Quantity of water in new mixture $= (105 \times 3) - 122 = 193l$

Therefore, ratio of water is to milk $= 193 : 122$

24 $A : B = 8 : 9$

$B : C = 2 : 3$

$C : D = 9 : 13$

$A : B : C : D = 144x : 162x : 243x : 351x$

But we need not to solve this, since we already know that

$B : C = 2x : 3x$

$\therefore 2x : 3x :: 18 : k \Rightarrow k = 27$ years

25

Milk		Water	
$5x$		$3x$	(initially)
$-25l$		$-15l$	
$5x - 25$		$3x - 15$	(after selling $40l$ mixture)
		$+15l$	
$5x - 25$		$3x$	(after adding $15l$ water)
5	:	4	

$\therefore \frac{5x - 25}{3x} = \frac{5}{4} \Rightarrow x = 20$

$\therefore$ New quantity of mixture $= (5x - 25) + 3x = 135$ litre

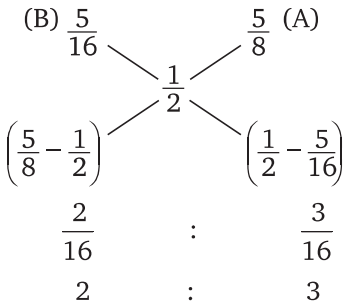
26

A	B
$C : T$	$C : T$
5 : 3	5 : 11

Concentration of copper in $A = \frac{5}{8}$

Concentration of copper in $B = \frac{5}{16}$

By alligation



So, the required ratio of $A : B = 3 : 2$ (Since $B : A = 2 : 3$)

27 $k + 10x = 6000$

$k + 25x = 9000 \quad (\because 25 \times 360 = 9000)$

$\Rightarrow \quad 15x = 3000$

$\Rightarrow \quad x = 200 \quad \text{and} \quad k = 4000$

$\therefore \quad k + 40x = 4000 + 40 \times 200 = 12{,}000$

Where k is the fixed expenditure.

28 Third proportional of 21 and 42 is 84 and mean proportional to 16 and 49 is 28.

Therefore required ratio $= \frac{84}{28} = \frac{3}{1}$ or 3 : 1

29 $P \propto \sqrt{l} \quad \Rightarrow \quad P = k\sqrt{l}$

$\therefore \quad 52 = k\sqrt{16} \Rightarrow k = 13$

Again, $P = k\sqrt{l}$

$65 = 13\sqrt{l}$

$\Rightarrow \quad l = 25\,\text{cm}$

30 $\frac{m+n}{m-n} = \frac{7x}{x} \quad \Rightarrow \quad \frac{m}{n} = \frac{4x}{3x}$

Again $mn = 4x \times 3x = 12x^2$

and $mn = 60x$

So $60x = 12x^2 \Rightarrow x = 5$

$\therefore \quad m = 20 \quad \text{and} \quad n = 15$

Hence, $\frac{1}{m} : \frac{1}{n} = \frac{1}{20} : \frac{1}{15} = 3 : 4$

Alternatively $\frac{m}{n} = \frac{4x}{3x}; \frac{1}{m} : \frac{1}{n} = 3 : 4$

31 $\underbrace{A+B}_{2} : \underbrace{C+D}_{1}$

and $\underbrace{B}_{3} : \underbrace{C}_{1} \quad \text{and} \quad \underbrace{C}_{3} : \underbrace{D}_{5}$

$\Rightarrow \quad B : C : D$

$9 : 3 : 5$

Again $A + B : C + D$

$\underbrace{7+9}_{16} \quad \underbrace{3+5}_{8}$

Thus $A + B = 16 \quad \Rightarrow \quad A = 7$ when $B = 9$

$\therefore$ Therefore share of $A = \frac{7}{24} \times 9600 = ₹\,280$

32 The best way is to consider some values in the proportion and verify the option, but care should be taken that in this type of question you are required to verify a possible correct option atleast with 2-3 times with different values.

Since, if you consider $p : q :: r : s \quad \Rightarrow \quad 1 : 2 :: 3 : 6$

you will find answer (b) is also correct, but for only particular values. So it is not the general solution. Finally you will realize that option (d) is the most appropriate answer.

33 Solve by componendo and dividendo or as mentioned in the previous solution.

34 Since for the constant distance time is inversely proportional to the speed. So, the required ratio of time taken by each of the rickshaw, car and scooter is

$\frac{1}{3} : \frac{1}{5} : \frac{1}{6} \quad \Rightarrow \quad 10 : 6 : 5$

$\left[\frac{1}{3} \times \frac{10}{10} : \frac{1}{5} \times \frac{6}{6} : \frac{1}{6} \times \frac{5}{5} = \frac{10}{30} : \frac{6}{30} : \frac{5}{30} = 10 : 6 : 5\right]$

35 $A : B = 2 : 3 \quad \text{and} \quad B : C = 3 : 5$

$\Rightarrow \quad A : B : C = 2 : 3 : 5$

$\therefore \quad (A + B) : C = 5 : 5 = 1 : 1$

Hence, Share of $(A + B) = \frac{1}{2} \times 6940 = 3470$

36 $x + y = 93 \quad \ldots(\text{i})$

and $\frac{x+3}{y+2} = \frac{24}{25}$

$\Rightarrow \quad 25x - 24y = (-27) \quad \ldots(\text{ii})$

From eqs. number (i) and (ii)

$x = 45 \quad \text{and} \quad y = 48$

Alternatively Go through options.

37 $\frac{a}{b} = \frac{4x}{9x} \quad \text{and} \quad \frac{a+4}{b+4} = \frac{21}{46}$

$\Rightarrow \quad a = 80 \quad \text{and} \quad b = 180$

$\therefore \quad b - a = 100$

38 $\frac{R}{D} = \frac{3x}{5x} \quad \text{and} \quad \frac{R}{D} \Rightarrow \frac{3x+10}{5x+10} = \frac{5}{7} \Rightarrow x = 5$

Hence, $\frac{R}{D} = \frac{15}{25}$

Alternatively Go through options.

39 Let the fraction be $\frac{x}{y}$ then $\frac{x+5}{y+5} = \frac{11}{15} \Rightarrow 11y - 15x = 20$

Since, we have only one equation in two variables, so we cannot find the solution.

40 $\frac{2x + 3x + 5x + 8x + 9x}{5} = \frac{162}{5} = 32.4$

41 The number of days required by a single Kirlosker pump to fill the tank $= 6 \times 7 = 42$ days and the number of days required by a single USHA pump to fill the same tanks $= 2 \times 18 = 36$ days. Now, since efficiency is inversely proportional to the number of days. Hence,

$\frac{\text{Efficiency of one K-pump}}{\text{Efficiency of U-pump}} = \frac{36}{42} = \frac{6}{7}$

42 $\frac{1}{5}$th field can be reaped by 96 man days

$\therefore \frac{4}{5}$th field can be reaped by $96 \times 4 = 384$ man days

Now, since there are only 8 days so reqired

$$\text{number of men} = \frac{384}{8} = 48 \text{ men}$$

43 $$210 = 2 \times 3 \times 5 \times 7$$

Therefore, $$\frac{N_1}{N_2} = \frac{2k \times 5 \times 7}{3k \times 5 \times 7}$$

Since, $$N_1 : N_2 = 2 : 3$$

Therefore, $N_1 = 70$ and $N_2 = 105$

$\therefore$ $$N_1 + N_2 = 70 + 105 = 175$$

44 $\frac{(53-x)}{(21-x)} = \frac{(41-x)}{(17-x)} \Rightarrow x = 5$

Alternatively Go through options.

45 $2x + 3x + 4x = 180 \Rightarrow x = 20 \quad \therefore \quad 4x = 80$

46 $(25 \times 12x) + (50 \times 4x) + (100 \times 3x) = 800x = 60000$

$\Rightarrow$ $$x = 75$$

$\therefore$ Number of coins of 25 paise $= 12x = 12 \times 75 = 900$

Alternatively Go through options, choices (a), (b) and (c) are eliminated since neither of 200, 225, 275 is divisible by 12. Hence choice (d) is correct.

47 Go through options.

Alternatively $\frac{(x-a)}{11} = \frac{(x-b)}{9} = \frac{(x-c)}{5} = k$

$\Rightarrow$ $$a = x - 11k$$
$$b = x - 9k$$
$$c = x - 5k$$

$\therefore$ $$x = \frac{(x-11k) + (x-9k) + (x-5k)}{2}$$

$$2x = 3x - 25k \Rightarrow x = 25k$$

$$a = 14k \Rightarrow b = 16k \text{ and } c = 20k$$

$\therefore$ $$a : b : c = 14 : 16 : 20 = 7 : 8 : 10.$$

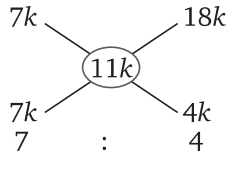
48 If the weight of kerosene be k kg/unit volume, then

Weight of petrol $= 7k$/unit volume

Weight of castrol $= 18k$/unit volume

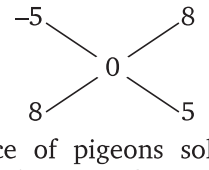
Required weight of the mixture $= 11k$/unit volume

By Alligation



The required ratio is 7 : 4.

49 **By Alligation**



The ratio of price of pigeons sold at loss and profit respectively are in the ratio of 8 : 5

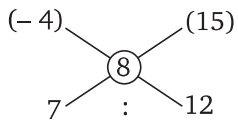
$\therefore$ Cost price of profitable pigeon $= \frac{5}{8+5} \times 182 = ₹ 70.$

50 $$\frac{C}{R} = \frac{3x}{5x}$$

Again $$\frac{C}{R} = \frac{3x \times 3}{(5x + 100)} = \frac{4}{5} \Rightarrow x = 16$$

$\therefore$ Price of Rotomac pen in 2000 was ₹ 80.

51 **By Alligation**



Now, we get the ratio of no. of rings sold at a loss and profit is 7 : 12.

$\therefore$ Number of rings sold at profit $= \frac{12}{19} \times 361 = 228$

Alternatively Go through options.

52 Best way is to go through options.

Consider option (d) $\frac{180}{30} + \frac{70}{35} = 6 + 2 = 8$ hrs

Hence, (d) is correct.

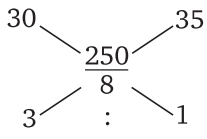
Alternatively $\frac{x}{30} + \frac{250 - x}{35} = 8$

$\Rightarrow$ $$x = 180 \text{ miles}$$

Alternatively Since the Average Speed

$$= \frac{\text{Total Distance}}{\text{Total Time}} = \frac{250}{8} \text{ miles/hr}$$

$\therefore$ **By Alligation**



Therefore, the ratio of time taken @ 30 m/hr and @ 35 m/hr is in the ratio of 3 : 1. It means he has travelled @ 30 m/hr for 6 hr.

Therefore, the distance travelled by car is 180.

($\because$ Distance = Speed $\times$ Time).

53 $\frac{22}{22} : \frac{17}{17} = 1 : 1$

54 $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{6}$

$\Rightarrow$ $\frac{20}{60} : \frac{15}{16} : \frac{12}{60} : \frac{10}{60}$ (by taking LCM)

$\Rightarrow$ $$20 : 15 : 12 : 10$$

$\therefore$ $$\text{Largest share} = \frac{20}{(20 + 15 + 12 + 10)} \times 171$$

$$= \frac{20}{57} \times 171 = 60.$$

55 Let the present age of Karishma and Babita be x and y then

$$\frac{x - 10}{y - 10} = \frac{1}{3} \quad \ldots(i)$$

Again $$\frac{x + 14}{y + 14} = \frac{5}{9} \quad \ldots(ii)$$

By solving eqs. (i) and (ii) we get $x = 26$ and $y = 58$

56 Let the fraction be $\frac{k}{5k}$, then $\frac{k+x}{5k+5x}=\frac{(k+x)}{5(k+x)}=\frac{1}{5}$

57 $$x \propto y \quad \text{and} \quad x \propto \frac{1}{z^2}$$

$$x \propto \frac{y}{z^2} \Rightarrow x = k\frac{y}{z^2}$$

$$6 = k \times \frac{75}{(5)^2} \Rightarrow k = 2$$

Again $$x = 2 \times \frac{24}{(4)^2} \Rightarrow x = 3$$

58 $$x \propto (y^2 + z^2) \Rightarrow x = k(y^2 + z^2)$$

$$15 = k(1^2 + 2^2) \Rightarrow k = 3$$

Again $$39 = 3 \times (2^2 + z^2)$$

$\Rightarrow$ $$13 = (4 + z^2) \Rightarrow z = 3$$

59 $$W \propto HA \Rightarrow W = K \times H \times A$$

Now, $$48 = K \times 1.2 \times 20 \Rightarrow K = 2$$

Again $$W = 2 \times 1.5 \times 30$$

$\therefore$ $$W = 90$$

60 $$(a+b):(b+c):(c+a) = 5x : 6x : 9x$$

And $$a + b + c = 10$$

$\Rightarrow$ $$(a+b)+(b+c)+(c+a) = 20x$$

$\Rightarrow$ $$a + b + c = 10x$$

$\Rightarrow$ $$a + b + c = 10 \qquad (\because x = 1)$$

$\therefore$ $$c = (a+b+c) - (a+b)$$

or $$c = 10 - 5 = 5$$

61

A	B	C	
$3x$	$4x$	$5x$	
$(3x + x)$	$2x$	$(5x + x)$	(B gives $\frac{1}{4}$ to A and $\frac{1}{4}$ to C)
$= 4x$	$2x$	$6x$	
$(4x + x)$	$2x$	$(5x)$	(C gives $\frac{1}{6}$ to A)
$5x$	$2x$	$5x$	
5	2	5	

☑ Level 02 *Higher Level Exercise*

1

M_1	M_2
$D : S$	$D : S$
$7 : 3$	$5 : 4$

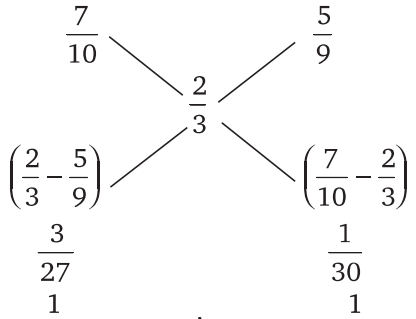
New Mixture	
D	S
2 :	1

Proportion of Dew in $M_1 = \frac{7}{10}$

Proportion of Dew in $M_2 = \frac{5}{9}$

Proportion of new mixture $= \frac{2}{3}$ (required)

By alligation rule



or $\frac{1}{9} : \frac{1}{30}$

or $10 : 3$

2 The ratio of quantities $= 140 : 60 = 7 : 3$

$\therefore$ The quantity to be exchanged $= \frac{(140 \times 3) + (60 \times 7)}{2(7+3)}$

$= 42$ litre

NOTE In this question individual prices are not required. You can solve it by forming equation.

Alternatively Go through options by assuming some prices.

3 Let the number be $100x + 10y + z$, then

$x + z = 6$ and $y + z = 5$

Also, from the given options only option (b) is suitable

i.e., $y + z = 5$ or $3 + 2 = 5$

4

Water	Pulp
$4x$	x
$3y$	$2y$

But $x = 2y$ since pulp remains contant

	Water	Pulp
$5y$ = loss in water $\rightarrow$	$8y$	$2y$
	$3y$	$2y$

But $10y = (8y + 2y) = 1.5$ kg

$\therefore$ $5y = (3y + 2y) = 0.75$ kg

5

Cost price	:	Selling price
$\frac{1}{9}$	:	$\frac{1}{6}$
2	:	3

or CP : SP = 2 : 3

Since profit = SP − CP

$\therefore$ Profit = 3 − 2 = 1

$\therefore$ Profit : CP = 1 : 2

6 Flow Chart :

